IOTAFLOW
SYSTEMS PVT. LTD

# ULTRASONIC BTU METER MANUAL

*" If You Want To Manage It, Measure It "*

www.iotaflow.com

# INTRODUCTION

This document covers battery powered ultrasonic flow meter **"IOTAFLOW ROBUST"** of ROBUST version and contains the information about its design, operation and installation.

Due to continuous improvement of product policy actual flow meter's specifications may differ from the data specified in this manual. However, this will not affect the metrological characteristics and functionality.

## LIST OF ABBREVIATIONS

| | |
|---|---|
| CP | - Primary converter |
| SMC | - Secondary measuring converter |
| DN | - Nominal diameter |
| ER | - Error |
| LCD | - Liquid crystal display |
| PC | - Personal computer |
| PS | - Pipeline section |
| PEA | - Electro-acoustic converter (transducer) |
| SPS | - Secondary power source |
| USS | - Ultrasonic signal |

NOTE. Words in the text marked in bold, for example, **Low**, correspond to the items displayed on the flow meter's screen or PC display when running Monitor ROBUST program.

# 1. DESCRIPTION AND OPERATION

## 1. Application

1. **"IOTAFLOW ROBUST"** battery powered ultrasonic flow meter of ROBUST version is designed to measure average volumetric flow rate and volume of bidirectional flows of cold water.

   The flow meters may be used as part of metering systems and complexes, in heat metering systems, automatic process control systems in power industry and municipal, etc.

2. **"IOTAFLOW ROBUST"** battery powered ultrasonic flow meter of ROBUST version performs the following functions:

- Measuring average volumetric flow rate in either forward or reverse flow directions
- Totalizing volume of forward and reverse flows independently or calculating their algebraic sum regarding flow direction
- Determining the current value of flow velocity and flow direction
- Output of measurement results to the frequency/pulse or logic output
- Output of measurement data on the indicator, as well as through the M-Bus interface or Wireless M-Bus interface
- Output of measuring, diagnostic, installation, archive and other information via RS-485 serial interface or NFC radio transponder according to ISO 15693
- Archiving of measurement results and setting parameters in energy-independent memory
- Possibility of entering setting parameters by means of software, considering individual features and characteristics of the measurement object
- Automatic control and indication of emergencies and failures, as well as recording of their type and duration in the archives
- Protection of the archival and adjusting data from unapproved access.

# 2. Specifications

1. Specifications of the flow meter **"IOTAFLOW ROBUST"** of ROBUST version are listed in Table 1.

**Table 1**

| Parameter | Value | Notes |
|---|---|---|
| 1. Nominal diameter of the pipeline, DN | 80 | |
| 2. Minimal measured average volumetric flow rate $Q_{min}$, $m^3/h$ | 0.25 | |
| 3. Maximal measured average volumetric flow rate $Q_{max}$, $m^3/h$ | 90.0 | |
| 4. Sensitivity threshold, $m^3/h$ | 0.057 | |
| 5. Maximal pressure in the pipeline, MPa | 1.6 | |
| 6. Properties of working fluid:<br>- temperature, °C<br>- density, kg/m3<br>- gas occlusion content, % of volume | <br>0 to 50<br>700 to 1200<br>no more than 5 | |
| 7. Power supply voltage, V | 3.6 | see clause 1.2.6 |
| 8. Mean time to failure, h | 75000 | |
| 9. Mean life time, years | 12 | |

2. The flow meter provides indication of the measured parameter values with the digit specified in clause 1.5.5.1 of this manual.
3. The flow meter provides that the results of measurements can be outputted to:
   - Universal output in the form of frequency, pulse or logic signal
   - logic output
   - RS-485 and M-BUS interface
   - Wireless M-Bus interface
   - NFC interface.
4. The flow meter provides storage of measurement results in the following logs:
   - Hourly log – 1440 records (hourly records) – 60 days
   - Daily log – 460 records (daily records)
   - Monthly log – 48 records (monthly records)
   - Mode log – up to 1000 record
   - Operating mode log – up to 4000 records.

   When powered off, the flow meter stores measurement results and configuration settings for no less than 1 year.
5. Environmental resistance:
   - To climatic influences – the ambient temperature range is from 5 to 50 °C, relative humidity is not more than 80% at temperatures up to 35 °C, without moisture condensation
   - To mechanical stress – vibration in the frequency range from 10 to 55 Hz with an amplitude of not more than 0.35 mm

- Atmospheric pressure ranges from 84.0 to 106.7 kPa.

The degree of protection of the flow meter corresponds to the IP67 or IP68. The required degree of protection of the device is determined when ordering.

6. The power supply of the flow meter is provided from an internal lithium C size battery with a nominal voltage of 3.6 V.
7. Appearance, overall dimensions and weight are given in Appendix A.

## 3. Metrological characteristics

1. The limits of permissible error in measuring the average volumetric flow rate, the volume of liquid in any direction of flow do not exceed the values given in table 2.

**Table 2**

| Error formula | Relative error limit value |
|---|---|
| $= 0.95 + \frac{0.1}{v}$ | 1.0 % at v > 2.00 m/s<br>1.5 % at v > 0.19 m/s<br>2.0 % at v > 0.10 m/s<br>3.0 % at v > 0.05 m/s<br>4.0 % at v > 0.033 m/s |

where v – flow velocity, m/s.

1.3.2. The diagram of real measurement error of the flow meter is shown in Fig.1.

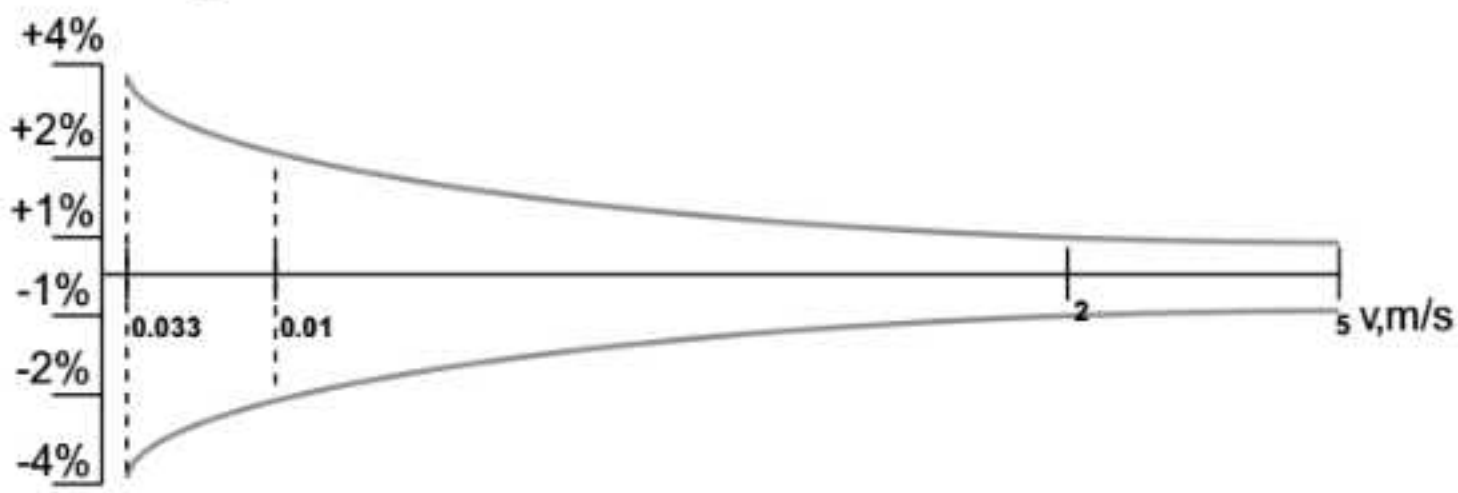



**Fig.1. Diagram of real measurement error of the flow meter**

## 1.4. Contents of the delivery package

Items of the delivery package are specified in Table 3.

**Table 3**

| Item | Qty | Notes |
|---|---|---|
| 1. Flow meter | 1 | Note 1 |
| 2. Installation kit | 1 | Note 2 |
| 3. Passport | 1 | |
| 4. Operation manual | 1 | Note 3 |

NOTES:

1. The size and the protection rating of the flow meter, the types of interfaces are set on request.
2. The required items of the delivery package are specified in the order sheet.
3. Operating documentation and order sheets for this product and other products are available on: www.iotaflow.com.
4. To connect the device with external devices via RS-485 interface, it is possible to supply a USB signal adapter – RS-232 / RS-485 on request.

# 5. Design and operation

## 1. Operation principle

1. The flow meter is designed on the base of a pulse-phase method of ultrasonic flow measurement. Its operation principle is based on measuring the time difference between the time it takes for an ultrasonic signal (USS) to travel through the liquid with the direction of flow and against it.
2. The peculiarity of this type of ultrasonic flow meter is alternating supply of electrical probing pulses generated by the secondary converter (SMC) to the converters PEA1 and PEA2 (Fig.2).

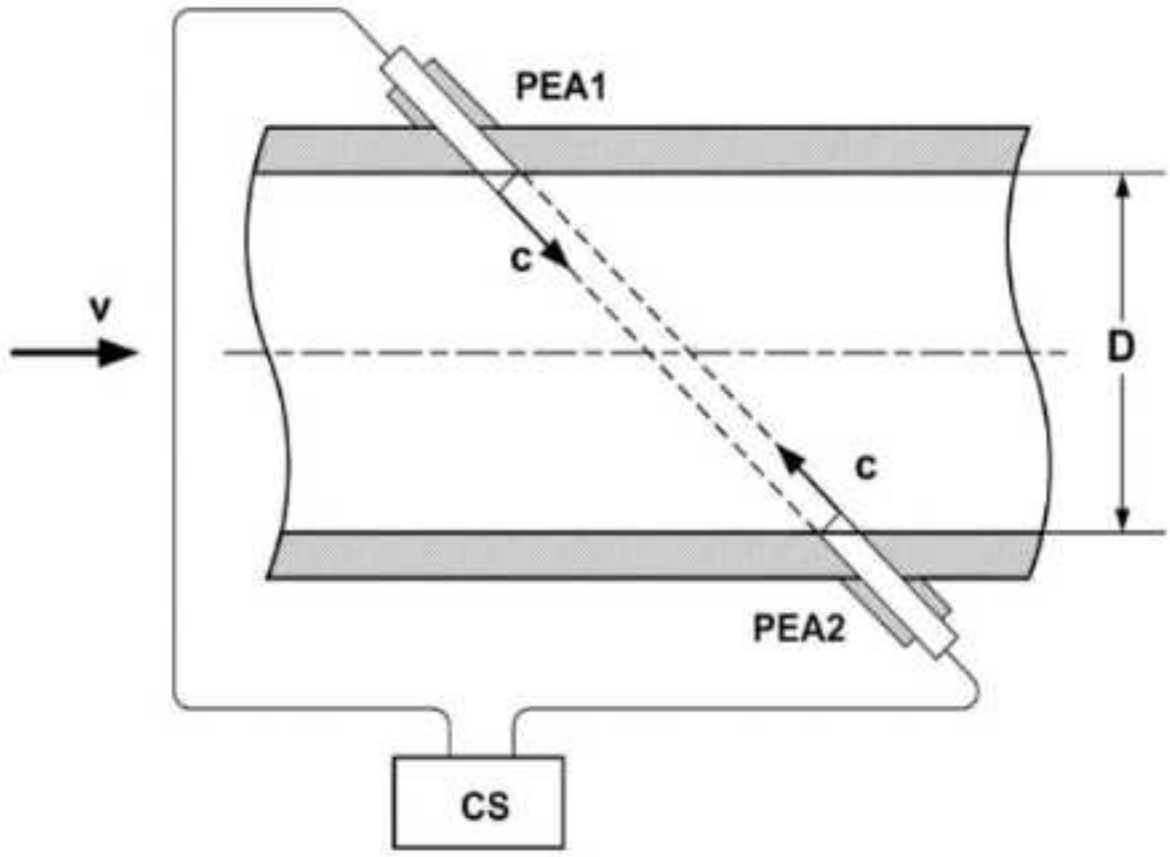



**Fig.2. Diagram of the pipeline section with PEA installed**

USS generated by one PEA is projected through the liquid flowing in the pipe and is received by the other PEA. Because of liquid flow, a sound wave is shifted, which in turn leads to the difference in upstream and downstream times of USS travel: it takes less time for an ultrasonic signal to travel in the flow direction (from PEA1 to PEA2) than that in the opposite direction (from PEA2 to PEA1). The difference in upstream and downstream time of USS travel through an acoustic path **dT** is proportional to flow velocity of fluid v and, correspondingly, to volumetric flow rate **Q**.

1.5.1.3. The flow rate value Q is calculated only if the following condition is fulfilled:

$$Q \geq Q_{min},$$

where $Q_{min}$ – minimal value of the flow rate at which the measured flow rate value in the flow meter is set to zero, the volume accumulation and pulse output at the universal output stops.

The recommended value of low flow cutoff corresponds to the flow meter's sensitivity threshold.

If $Q > Q_{max}$ condition (where $Q_{max}$ responds to the flow velocity value of 5 m/s) condition is fulfilled, flow measurement continues, but volume accumulation, archiving and pulse output at the universal output stops.

### 1.5.2. Design

The block diagram of flow meter **"IOTAFLOW ROBUST"** of ROBUST version is shown in Fig.3.

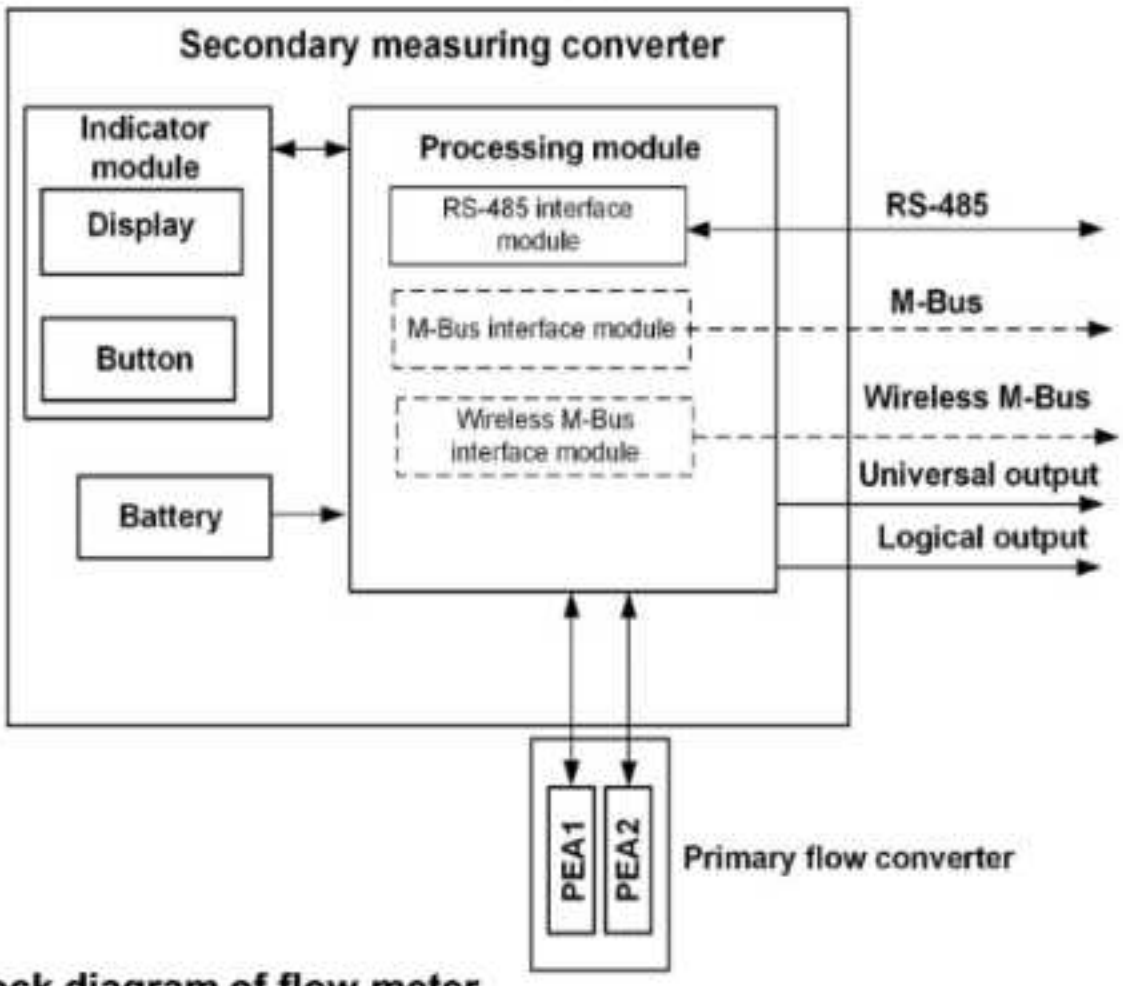



**Fig.3. Block diagram of flow meter**

The ROBUST flow meter consists of a primary converter (CP) and a secondary converter.

The primary flow converter is a pipeline section (pipe section with pipe connections for PEA installation and flanges for pipe connection) with two PEAs installed on it (Fig.A.1 of Appendix A).

The PEA operates in two modes: radiation, when the electrical pulse signal from the secondary converter is converted into ultrasonic vibrations, and reception, when the ultrasonic vibrations of the liquid are converted into an appropriate electrical signal.

The secondary converter contains a processing module that acts as a meter and calculator as well as a display module. The processing module controls the electro-acoustic sounding, processes the measurement signals from the PEA and performs the secondary processing of the measurement information and ensures the storage of the measurement results.

External communications of the flow meter are provided by means of universal and logic outputs available on the processing module as well as optional RS-485 serial interface modules, M-Bus interface or Wireless M-Bus interface installed in the processing module.

The measuring information is displayed on a graphical liquid crystal display (LCD) located on the display module. LCD provides output of two lines of alphanumeric information at 16 characters per line. The information output is controlled by means of an optical button located on the front panel next to the LCD.

Before commissioning, the flow meter is set up using a personal computer (PC) via RS-485 interface.

## 3. Access levels

1. The flow meter has three levels of access to setting and calibration parameters.

The access levels differ in the information displayed, possibilities to change the setting and calibration parameters of the flow meter and are referred to as WORK, SERVICE and ADJUSTMENT modes.

Mode applications:

- ADJUSTMENT – adjustment and verification mode;
- SERVICE – preparation for operation mode;
- WORK – operation mode (user mode).

Modes differ in the level of access to information and possibilities to change the settings of the flow meter via RS-485 interface.

Maximum rights are granted in ADJUSTMENT mode. The mode allows the user to view all parameters and to modify all configuration settings. Minimum rights are given in WORK mode.

The set of parameters displayed does not depend on the set operation mode.

2. The operation mode is defined by a combination of the presence/absence of a short-circuit using the contact pair jumpers J1 and J2 located on the processing module (see Fig.A.3 of Appendix A).

Correspondence of combinations to operation modes is resulted in table 4, where "+" is presence of contact pair closing, and "-" is absence of closing.

**Table 4**

| Operation mode | Contact pair | | Mode application |
|---|---|---|---|
| | J1 | J2 | |
| WORK | - | - | Operation |
| SERVICE | - | + | Preparation for operation |
| ADJUSTMENT | + | - | Adjustment |

3. WORK mode is a flow meter operation mode on the site.

WORK mode allows the user to view the data.

1. With the indicator: current date and time, values of measured parameters (volume flow rate, volumes accumulated during forward and reverse flow direction, as well as their algebraic sum), status line of emergencies and software version number.
2. With RS-485, M-Bus or Wireless M-Bus interfaces:

a) Values of measured parameters: volume flow rate, volumes accumulated in the forward and reverse flow direction, as well as their algebraic sum, fluid velocity, and current velocity of USS

3. With RS-485 interface:

a) The contents of the archives and event log

b) Configuration parameters: current date and time, characteristics of universal and logic outputs

c) Operation parameters:

- Signal processing characteristics (median averaging, arithmetic averaging, etc.)
- Communication parameters via RS-485 interface
- Types of emergency situations.

The WORK mode allows the user to change communication parameters via RS-485 interface (network address of the device, exchange rate from 1200 to 4800 baud).

4. SERVICE mode is a mode of preparation of the flow meter for operation on the site.

SERVICE mode additionally (in relation to WORK mode) allows via RS-485 serial interface to:

a) View:

- The configuration of the pipeline section (base, axial base, nominal diameter)
- Setting parameters for the ultrasonic signal
- Zero offset value and additional USS delay

b) Change:

- Operation modes and parameters of the universal and logic outputs
- Flow and volume measurement units ($m^3$/h; l/min, $m^3$, l)
- Device clock readings
- Device clock transition mode to "winter"/"summer" time

c) Adjust LCD contrast, operating time and measurement information update period

d) Start the calculation of conversion factor or pulse weight for the universal output

e) Zero accumulated volumes and archives.

1.5.3.5. ADJUSTMENT mode allows to view and modify all parameters without exception.

Also, ADJUSTMENT mode allows to record the factory number of the device via RS-485 serial interface.

## 4. Recording of operation results

1. The results of measurements and calculations are recorded in the internal archives: hourly, daily, monthly, with the depth of the archives, given in clause 1.2.4. Access to the archives is carried out through the RS-485, M-Bus or Wireless M-Bus interfaces and NFC radio transponder.

Hourly, daily and monthly archives are cyclical and have the same structure.

The values of the following parameters are recorded in each record:

- **V+** – total volume in the direct flow direction over the archiving interval, $m^3$ (l)
- **V-** – total volume in the reverse flow direction over the archiving interval, $m^3$ (l)
- **ER** – error code
- **ER time** – total time, during which there was no accumulation of volumes in the archive

- **Worktime** – operating time for the archiving interval.

2. Changes in the operating mode of the device are recorded in the mode log, which can contain up to 1000 records.

   Each mode log entry records:

- Sequence number of the record
- Name of the set operating mode of the device
- Date and time of mode setting.

## 5. Information output

1. Liquid crystal indicator

   The list of parameters, which are displayed on the flow meter's indicator, is shown in Table 5.

**Table 5**

| Designation | Parameter | Measurement units (format) | Number of display symbols | |
|---|---|---|---|---|
| | | | integral part | fractional part |
| **Time** | Current time | **= XX:XX** | | |
| **Data** | Current date | **XX.XX.XX** | | |
| **Q** | Current volume flow rate regarding flow direction | **$m^3$/h, l/min** | up to 4 | 4 |
| **V+** | Direct flow volume (cumulative total) | **$m^3$, l** | up to 9 | 4 |
| **V-** | Reverse flow volume (cumulative total) | **$m^3$, l** | up to 9 | 4 |
| **V** | Total volume (cumulative total) | **$m^3$, l** | up to 9 | 4 |
| **Err** | Status bar | **– – – – – –** | | |
| **Tw** | Total operating time | **XXX:XX h:m** | 3 | 2 |
| **Tner** | Fault-free operation time | **XXX:XX h:m** | 3 | 2 |
| **K1** | Calibration factors | **X.XXXXXX** | 1 | 6 |
| **P1** | | **-X.XXXXXX** | | |
| **K2** | | **X.XXXXXX** | | |
| **P2** | | **-X.XXXXXX** | | |
| **Soft** | Software version number | **XX.XX.XX.XX** | | |
| **CRC** | Software checksum | **0xA8D9** | | |

NOTES:

1. The flow rate in the reverse flow direction as well as the negative values of the total volume and the volume of the reverse flow are indicated with a minus.
2. The total volume is defined as the sum of the volumes accumulated in the forward (positive) and reverse (negative) flow direction, considering the sign of the flow direction.
3. An overflow of counters occurs if **V** 2 $10^9$ $m^3$. After the counter is full, the display continues from zero.

   In the SERVICE and ADJUSTMENT modes, the indicator is always on. In the WORK mode, the indication is switched on by pressing the button and is switched off after the last pressing it in the interval set in the **Worktime** field in the **Service** tab of the Monitor ROBUST application.

   Switching the indication of the parameters given in Table 5 is performed on the ring by sequential pressing the button located on the front panel of the device.

NOTE. Using the optical button in the flow meters has the following features:

- The button works "slowly", i.e. its touching should last more than 1 s
- The button is triggered when the finger is taken off.

1.5.5.2. RS-485 interface

RS-485 serial interface allows to read out measuring, archival, installation and diagnostic information, to modify setting parameters. RS-485 interface supports ModBus protocol (RTU ModBus and ASCII ModBus)

RS-485 interface provides cable communication in a group of several subscribers, one of which can be a PC, with a communication line length of up to 25 m.

At the time of release from production the network address is recorded in the flow meter – 1. Using switches SK1 and SK2 on the interface module (see Fig.A.3), you can change the network address of the device according to Table 6.

**Table 6**

| Switch position | | Network address |
|---|---|---|
| SK1 | SK2 | |
| 1 | 2 | 2 |
| 2 | 1 | 3 |
| 1 | 1 | 4 |

When both switches are set to position 2 on the interface, you can set any network address of the device (from 1 to 255).

The RS-485 baud rate and other communication parameters are set programmatically.

3. M-Bus interface

The M-Bus interface complies with EN 13757. The interface allows the readout of measured data, such as: current flow rate, accumulated direct, reverse and total volumes, operating time, as well as the presence of emergencies under the numbers 1 ... 3 (see Table 7).

By default, the flow meter has a network address of 1 and an exchange rate of 2400 baud. The interface is connected via two wires, the polarity of the connection is not important. The M-Bus interface baud rate is set programmatically.

**WARNING:** Power supply of the flow meter and M-Bus modem from one power source is prohibited.

4. Wireless M-Bus interface

The Wireless M-Bus interface complies with EN 13757-4. The interface allows the transmission of measured and archived data to a receiver (WM-Bus dongle interface box, hub, etc.).

T1 mode is implemented in the protocol: the flow meter works only for transmission without requests from the data acquisition device. The transmission period is 16 seconds and the carrier frequency is 868 MHz.

Each parcel transmits a Wireless M-Bus identifier, which includes the serial number of the flow meter, which determines which metering device (from which consumer/subscriber) received the data.

5. Universal output

The flow meter has a non-isolated universal output that can operate in frequency, pulse and logic modes.

The functions of the universal output, operating modes, output parameters and output disconnection are defined by the software settings.

The diagram of the output terminal stage and the description of its operation are given in Appendix B.

◆ In the frequency mode, a pulse sequence of meander type with a duty cycle of 2 is given to the open output, the frequency of which is proportional to the current value of the flow rate. It is possible to scale the frequency output operation by software setting the **Maximum output frequency** or **Output conversion factor**, as well as the lower and upper flow limit values of **Lower limit by flowrate** and **Upper limit by flowrate**, corresponding to the sequence frequency of 0 Hz and the maximum frequency, respectively. The maximum possible value of **Maximum output frequency** is 1000 Hz in ADJUSTMENT mode and 10 Hz in WORK and SERVICE modes.

**Error frequency** is pulse sequence frequency (not more than 10 Hz), which will be formed at the output if the measured value of the flow rate exceeds the value of $Q_{max}$ for this DN flow meter. The target value of **Error frequency** must be at least equal to the target value of **Maximum output frequency**. To disable the alarm frequency output function, set **Error frequency** to 0 Hz.

◆ In pulsed operation mode, a pulse burst is given to the open output every second, the number of which, considering **Pulse weigt**, corresponds to the volume value measured in the previous second. The maximum possible pulse repetition rate in the burst (of meander type with a duty cycle of 2) is 1000 Hz in ADJUSTMENT mode and 10 Hz in WORK and SERVICE modes.

◆ For correct operation of the flow meter's universal output there is a procedure of automatic calculation of **KP** parameter (pulse/$m^3$) in frequency mode and **Pulse weigt parameter** ($m^3$/pulse) in pulse mode.

**KP** parameter is calculated based on user-defined values of **Q lower limit**, **Q upper limit** and **Fmax**, **Pulse weigt** parameter is calculated on the basis of defined values of **Q upper limit** and **Pulse width** in the range:

- From 1 to 1000 ms – in ADJUSTMENT mode
- From 100 to 1000 ms – in WORK and SERVICE mode.

◆ In the logic mode, the presence of an event (or its certain condition) at the output corresponds to one level of electrical signal at the output, and the absence of an event (or its other condition) – to another level of signal.

The active signal level (**Active level**), i.e. the level of the signal corresponding to the presence of an impulse (event), is set by setting the **High** or **Low** value in any operation mode. The electrical parameters of the signal levels are given in Appendix B.

When released from production, the typical values of universal output parameters are set: pulse type, passive mode, pulse weight – 0.0025 $m^3$/pulse.

1.5.5.6. Logic output

The flow meter has a non-isolated logic output. Operation modes, parameters and electrical characteristics of the logic output are similar to those of the operation modes, parameters and electrical characteristics of the universal output in the logic mode.

# 6. Flow meter design

1. The primary flow converter and the secondary measuring converter form a single design of the flow meter (see Fig.A.1 of Appendix A).

   The pipeline section (PS) of the primary converter is made of high-strength plastic PA66-GF30. The ends of the PS are fitted with pipe flanges for connection to the counter flanges.
2. The plunge PEA used in the flow meter is cylindrical, the end of the PEA has a radiating plane in the form of a disk. The converters are installed in the pipeline section in such a way that the radiating PEA plane is in contact with the liquid to be monitored. The PEAs supplied with CP are designed for pressures up to 1.6 MPa.
3. PEA and the secondary converter are connected by communication cables hidden inside the enclosure.
4. The secondary converter is mounted directly on the flow meter's pipeline section. The processing module board is placed in the secondary converter enclosure, and the indication module board with indicator – in the front panel. The modules are connected to each other by a signal loop.
5. After installation of the interface cable and the universal output, the secondary flow meter converter with IP68 protection is completely filled with Hensel GH 0350 type compound. The design of the device is not disassembled.
6. The appearance of the flow meter may differ from that given in Appendix A without changing its operational and metrological characteristics.

# 6. Marking and sealing

1. The marking on the front panel of the SMC contains the designation and name of the flow meter, the manufacturer's trademark and the approval mark of the measuring device type. In addition, the maximum temperature of the liquid to be measured, the supply voltage and the size (DN) of the flow meter are marked on the individual labels.

   The factory numbers of the PEA and pipeline section are indicated on their enclosures.
2. After verification of the flow meter the contact pair of the permission to modify the calibration parameters is sealed. Also, the screws of fixing the protective cover of the processing module are sealed with the manufacturer's seal.
3. The contact pair of the permission to modify the service parameters can be sealed after the commissioning.

   In addition, the cover of the secondary converter can be sealed with two hinged seals to prevent unauthorized access during transport, storage or operation.

# 2. INTENDED USE

## 1. Operational restrictions

1. The flow meter should be operated under the conditions of influencing factors and parameters of the controlled medium, which do not exceed the permissible values specified in the operating documentation.
2. The flow meter can be installed in vertical, horizontal or inclined piping.
3. Precise and reliable operation of the flow meter is ensured if the following conditions are met at the CP installation site:
   - Fluid pressure in the pipeline and its operation modes exclude gas formation and/or accumulation of gas (air)
   - The internal volume of PS must be filled with liquid during operation
   - Sections of the pipeline up to the flow meter should not contain devices or structural elements that cause changes in the structure of the fluid flow.
4. Type and composition of the controlled liquid (presence and concentration of suspended solids, foreign liquids, etc.), operation mode and condition of the pipeline should not lead to corrosion and/or deposits that affect the performance and metrological characteristics of the flow meter.
5. Flow meter should be installed on the site in accordance with the section "Installation of the flow meter" of this manual.
6. Lightning protection of the device site protects the device from failure in the presence of lightning discharges.
7. The requirements for the operating conditions and the selection of the installation site given in these operating instructions take into account the most typical factors affecting the operation of the flow meter.

   Factors that cannot be predicted, evaluated or checked and which the manufacturer could not consider in the development of the flow meter may exist or occur during its operation.

   If such factors occur, they must be eliminated or a different place of operation must be found where these factors do not exist or do not affect the operation of the product.

## 2. Safety measures

1. The flow meter can be operated by the personnel who have studied the documentation for the product.

2. When operating the flow meter, dangerous factors are:
   - Pressure in the pipeline (up to 1.6 MPa)
   - Other dangerous factors related to the nature and characteristics of the facility where the flow meter is operated.

3. If external damage to the device or communication cables is detected, disconnect the device until a specialist determines that it can be used again.

4. During installation, commissioning or repair of the flow meter is prohibited to:
   - Make connections to the flow meter, switching operation modes when the power supply is on
   - Dismantle the flow meter on the pipeline until the pressure is completely relieved in the section of the pipeline where the work is performed
   - Use electrical devices and power tools without connecting their enclosures to the protective grounding line, as well as to use the listed devices in a faulty condition.

# 3. INSTALLATION OF THE FLOW METER

## 1. Transportation

The flow meter must be transported to the installation site in its original packaging.

After transportation to the place of installation at a negative temperature and introduction into the room with a positive temperature, the flow meter should be kept in the package at least 3 hours to avoid moisture condensation.

When unpacking the flow meter, check its completeness in accordance with the passport for this device.

## 2. Installation requirements for the flow meter

1. To install the device on the site it is necessary to have a free section in the pipeline for installation of the flow meter.

   The following conditions must be met at the CP installation site:

   - Liquid pressure and pipeline operation modes exclude gas formation and/or accumulation of gas (air)
   - The internal volume of the CP must be filled with liquid during operation;
   - there are no or minimum pulsations and swirls of liquid.

   CP may be installed in horizontal, vertical or inclined piping (Fig.4). The CP (PEA) must not be located at the highest point of the pipe section. The most suitable place for installation, if any, is the upstream or downstream section of the pipeline.

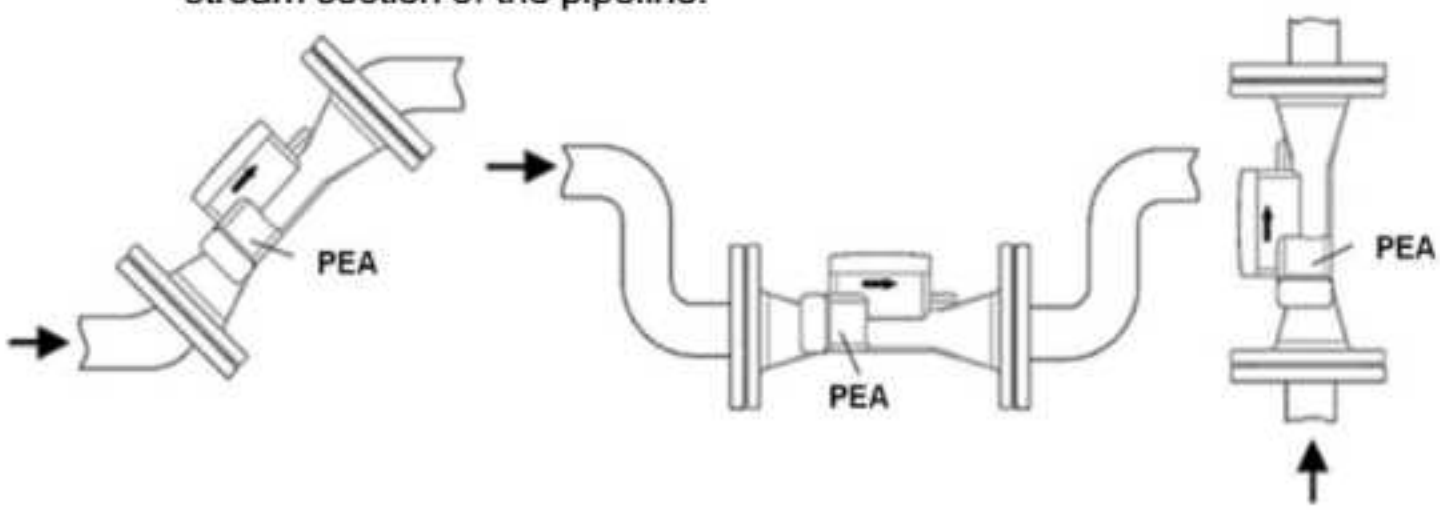



**Fig.4. Recommended installation locations for the CP of flow meter**

NOTE. If it is not possible to install the flow meter in the recommended places, it is possible to install the device at the top of the pipeline. It is necessary to install an air valve (air vent) at a point above the upper point of the flow meter, for example, in the expansion of the pipeline.

**WARNING:** Ensure that the air valve is in good working order. Correct operation of the flow meter is possible only when there is no air in the duct.

2. The difference between the inner diameters of the pipeline and the CP pipeline section at the jointing points must not exceed 0.05·DN.
3. When installing the CP on a pipeline, it is recommended that the longitudinal plane of PEA (the plane passing through a pair of PEA along the

pipe axis) be positioned at an angle of = 45 -90 to the vertical plane (Fig.5).

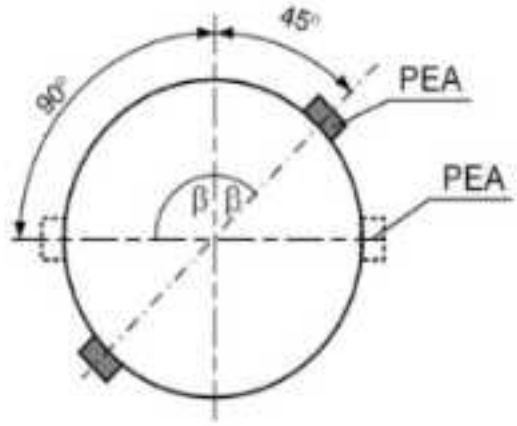



**Fig.5. Recommended PEA positions when installing the flow meter**

## 3. Requirements for the length of straight sections

1. If there are hydraulic resistances at the installation site, normal operation of the flow meter is provided by straight sections of the pipeline before the first and after the last one PEA in the stream. The minimum values of the relative length of straight sections for different types of hydraulic resistances are given in Table D.1 of Appendix D.

   The length of the straight section L (mm) is determined by the formula:

   $$L = N\ DN,$$

   where N is the relative length expressed by the number of DNs and shown in Table D.1;

   DN – nominal diameter of CP, mm.

   **WARNING:** When measuring the reverse flow rate, all PEAs are the first in the flow and the lengths of the straight sections must be determined from this position.

2. When installed in the pipeline in front of the primary flow straightener converter (Appendix D), the straight input length of the CP can be reduced by up to two times.

## 3. Installation on the pipeline

1. Before starting work on the pipeline in the place of installation of the flow meter, fix pipe sections that may deviate from the coaxial position after cutting the pipeline with clamps to the fixed supports.

   **WARNING:** Before installing the flow meter it is necessary to drain the liquid and cover the section of the pipeline, where the installation works will be carried out.

2. A section of the required length is cut out at the selected point of the liquid-free pipeline, and counter flanges of the appropriate diameter are welded to the pipe ends. Ensure the alignment and parallelism of the flanges, between which the CP of flow meter is installed.

3. The flow meter should be installed in the pipeline, and the direction of the arrow on the CP should coincide with the flow direction or the forward direction for reverse flow. The joints between the flanges are sealed with gaskets and the flanges are bolted together.

**WARNING:** Do not drop or hit the flow meter during installation. It may result in damage to the installed PEA or SMC.

**NEVER** remove the PEA from the mounting nozzles.

# 5. Electrical installation of the flow meter

1. The communication cable of the flow meter is connected to the SMC in the manufacturer. A round cable with an outer diameter from 3.0 to 6.5 mm is used as a communication cable.
2. External communication cable (SMCs are external devices) must be routed according to the operating conditions of the flow meter.
3. It is not recommended that the excess part of the cable be wound up with rings.
4. The external cable is attached to the wall as far as possible. It is recommended to place the cable in a metal or plastic pipe (hose) to protect it from mechanical damage.

**DO NOT** route the signal cable near power circuits or in the presence of high level electromagnetic interference (e.g. in the presence of a thyristor regulator) without laying it in an earthed steel pipe or metal hose.

**WARNING:** To ensure the declared degree of protection of the flow meter during installation, the following requirements must be met:

- The seal on the back of the SMC front panel must be clean and undamaged
- Before installing the front panel on the SMC enclosure, check that the seal is evenly positioned in the ring groove provided for it without tension or protrusions, and that it does not fall out when the front panel is turned upside down. Silicone sealant can be used to fix the sealant
- The SMC front panel should be tightened securely with screws after installation
- The cable gland packing nut must be tightened securely.

It is necessary for the signal cable to form a downward U-shaped loop in the vertical plane (Fig.6) to prevent dripping water or condensate from entering the SMC through the cable entry.

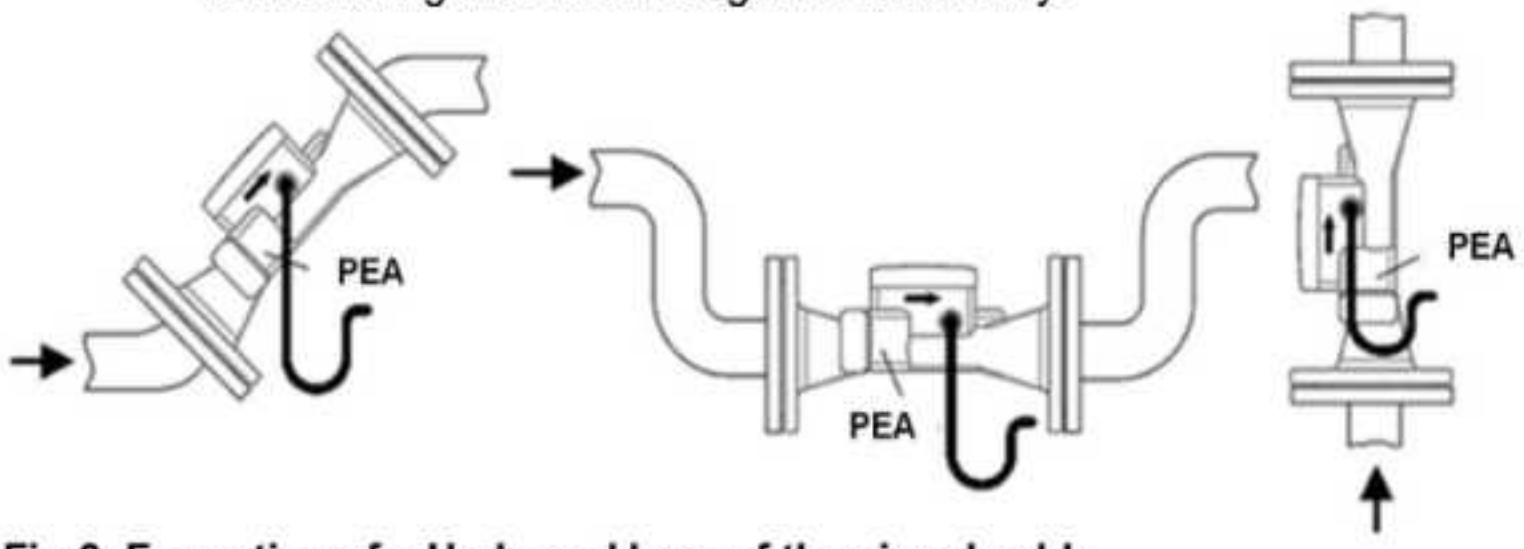


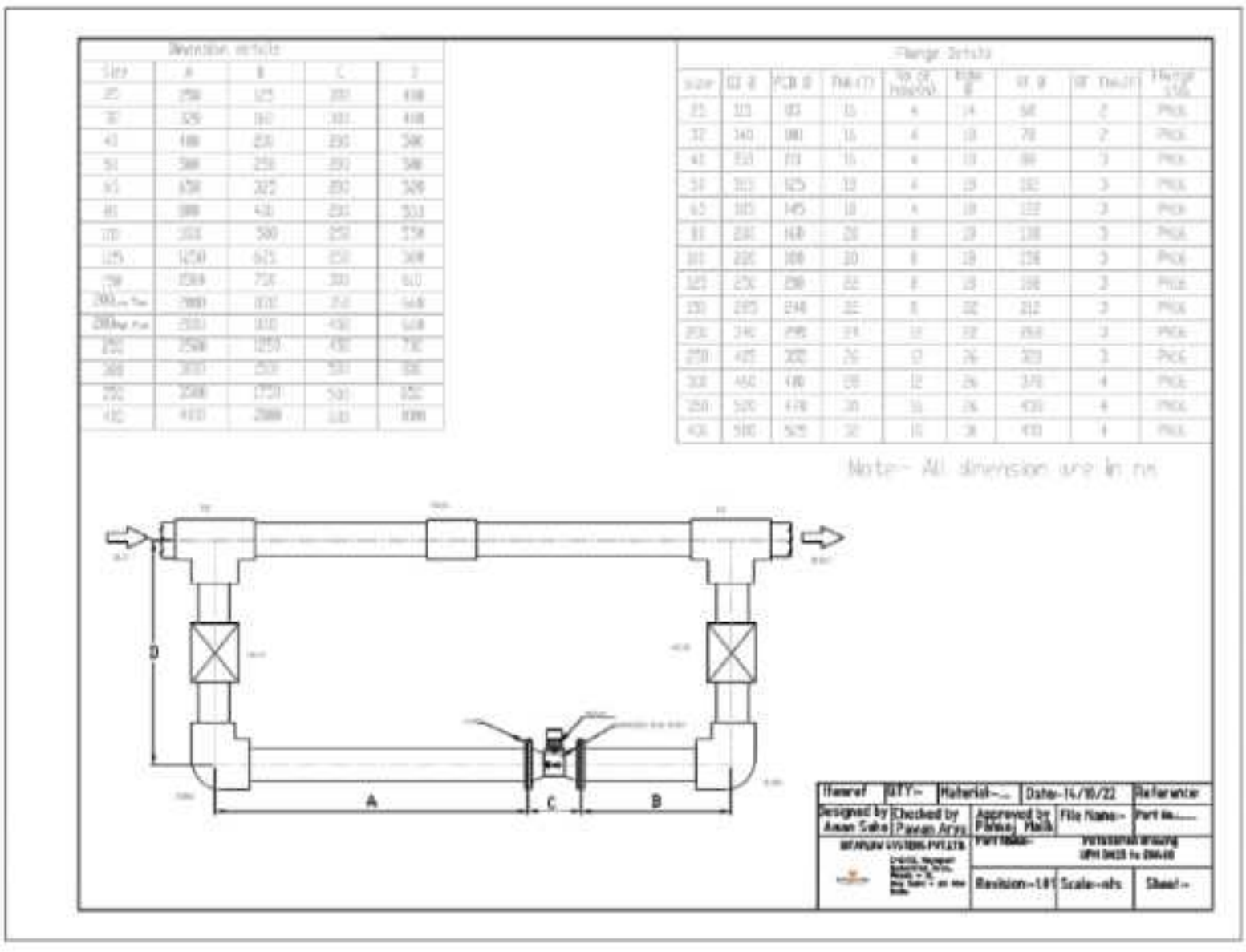

**Fig.6. Formation of a U-shaped loop of the signal cable**

**WARNING:** The manufacturer **SHALL NOT BE LIABLE UNDER WARRANTY** for failure to meet the requirements for the declared degree of protection and for leaks through the cable gland.

## 6. Dismantling the flow meter

When dismantling the flow meter:

- Shut off the liquid supply, make sure that there is no pressure in the pipe-line and drain the liquid
- Disconnect the communication cable suitable for the SMC
- Remove the flow meter.



# 4. OPERATION

## 1. Setting up before operation

1. The flow meter can be configured and read in the WORK and SERVICE modes via the serial interface using the Monitor ROBUST service application included in the UB Viewer ModBus program. The installation file can be downloaded from the delivery set.

   The UBViewer ModBus program works under Windows XP operating systems (Vista, Windows 7, Windows 8, Windows 10). For the installation of the program on the PC, open the installation file and follow the instructions appearing on the monitor.

2. Before starting the operation, cable the flow meter RS-output and the computer serial port (via RS-232/RS-485 signal adapter) or the computer USB port (via USB-RS-232/RS-485 signal adapter).

   Set the flow meter to SERVICE mode by setting a jumper on the corresponding contact pair. Start the "UBViewerProjects" file on the computer from the folder where it was loaded during installation. The UB-Viewer ModBus menu bar appears on the monitor. Open the **"IOTAFLOW ROBUST** .vpr" file, and the **Monitor ROBUST** application window will be displayed (Fig.7).

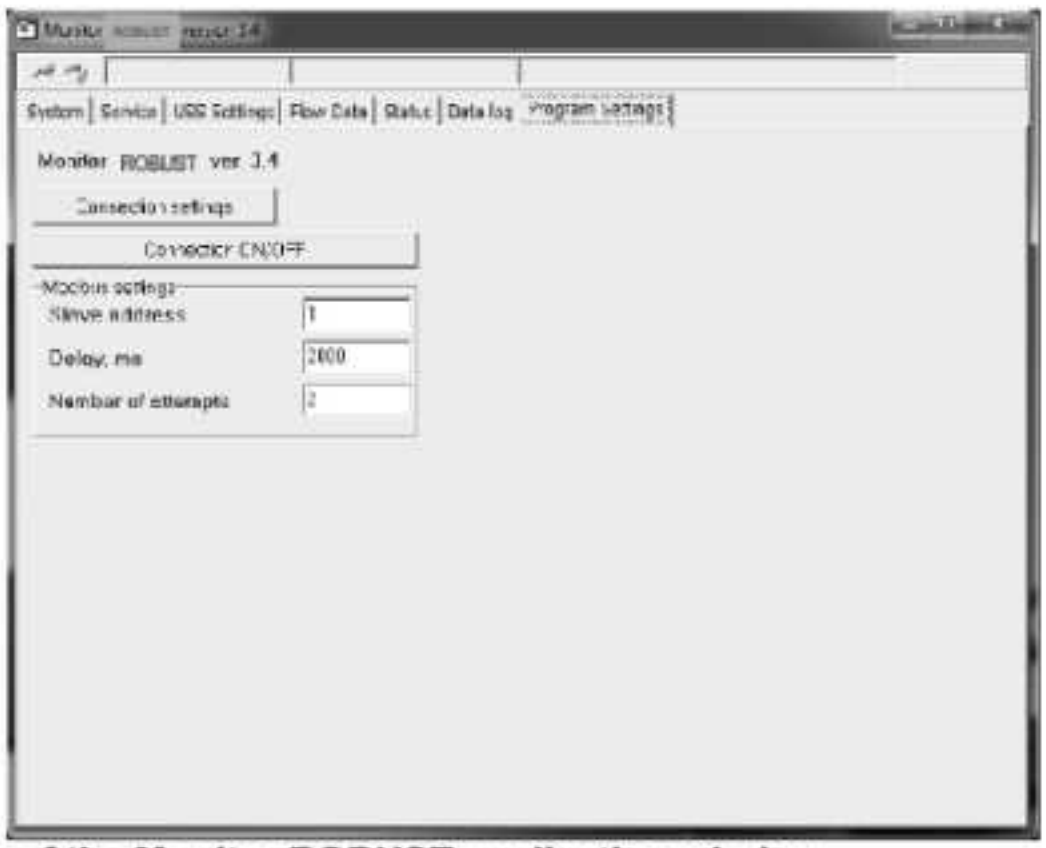



**Fig.7. View of the Monitor ROBUST application window**

The **Monitor ROBUST** application window interface consists of seven tabs divided by function. The contents of the tabs depend on the operating mode of the device. The operating mode is automatically determined when a PC communicates with the flow meter.

To set up the connection parameters, press **Connection settings** button on the **Programm Settings** tab or press the (Edit parameters of connection with the device) button in the upper left corner of the application window (Fig. 7).

Set the following in the appeared **Data link setting** window (Fig.8):

- The number of the PC COM port to which the signal adapter is connected
- Baud rate (4800 bps).

Other setting parameters are loaded automatically.

**Fig.8. View of the Data link setting window**

To set up communication with the flow meter, press the button in the upper left corner of the application window (Fig.7). If the operation is successfully completed, the line to the right of the button should contain the following messages: **"The data link is open"** and **"The command is executed successfully"**. The button will change the view to (Close communication channel). Click the **System** button in the command line, the **System** tab will appear in the application window (Fig.9).

**Fig.9. View of the System tab after the PC has been connected to the flow meter**

You can adjust the device time, set up communication parameters via the interface, reset archives and volumes, and set the mode of transition to "winter"/"summer" time on the **System** tab.

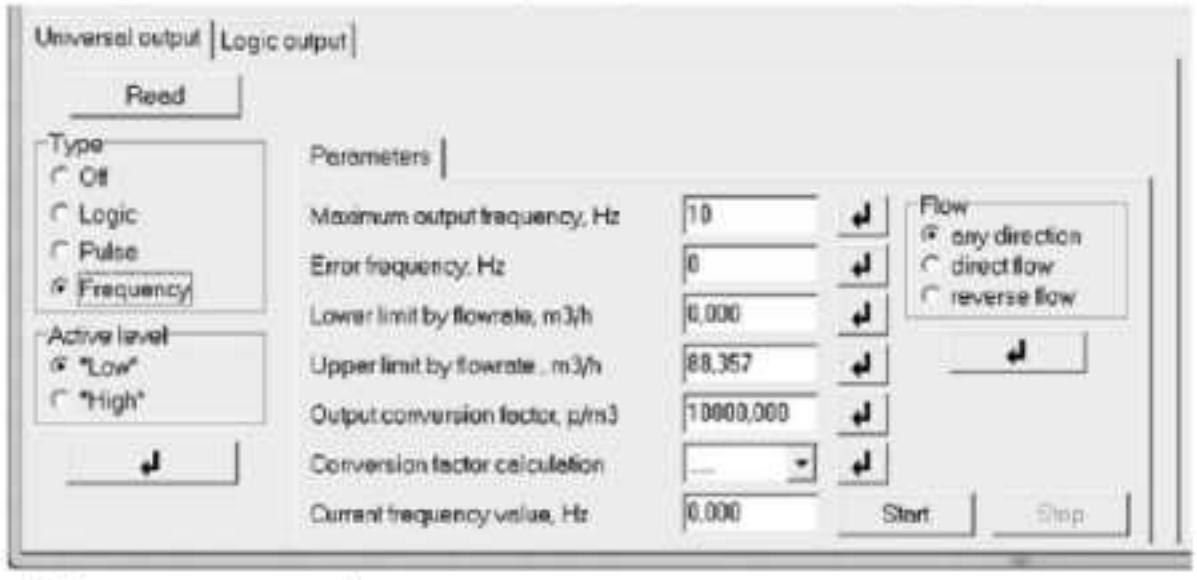
4.1.3. You can set the USS processing parameters, set the LCD contrast value and operating time and select the measured flow rate dimension on the **Service** tab (Fig.10).

**Fig.10. View of the Service tab**

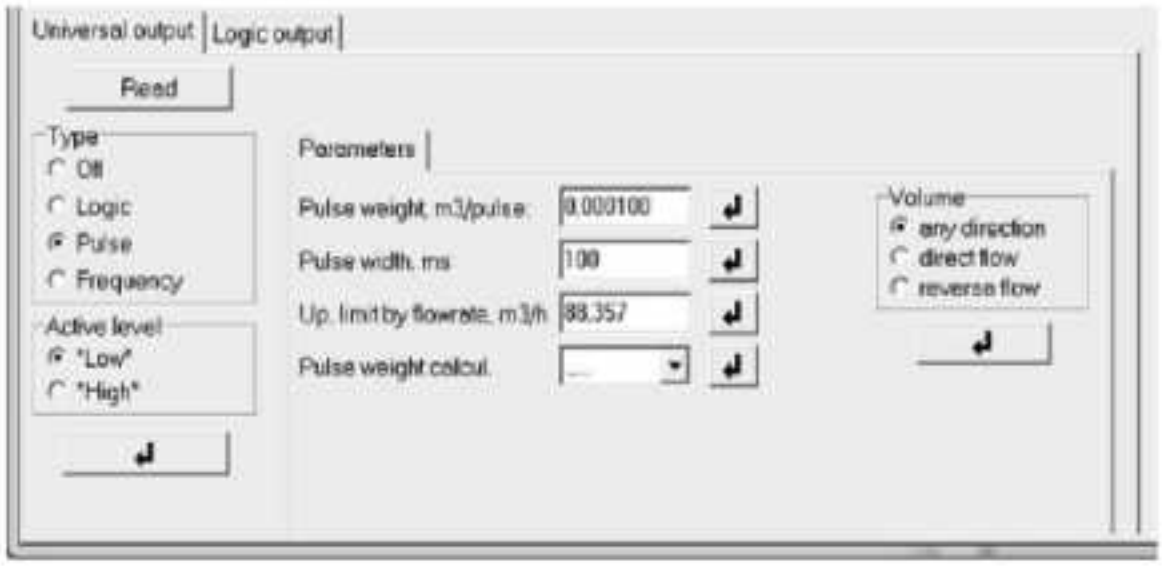
To change the current value of the parameter, type a numeric value in the corresponding field or set a symbolic value from the drop-down list and press the button.

You can also set the parameters of the universal and logic outputs in the **Service** tab.

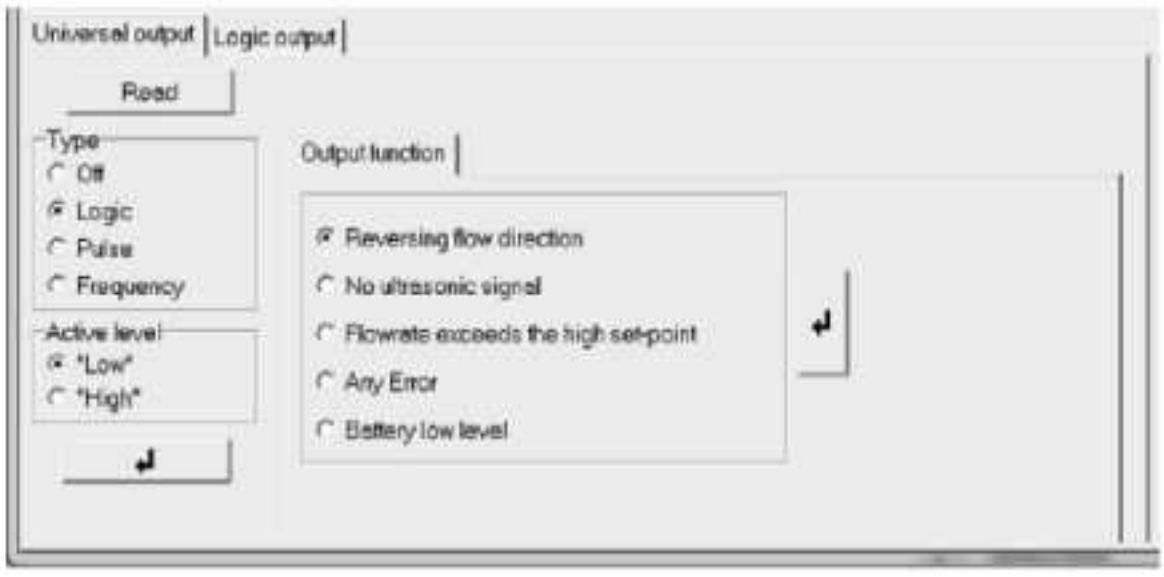
To switch to the setting parameters of universal output operation, set the checkbox ◉ in the additional tab **Universal output** in the line containing the name of the output operation mode (Fig.11).




a) Frequency mode




b) Pulse mode




c) Logic mode

**Fig. 11. Configuration parameters for different universal output modes**

Configuration parameters of the logic output operation are contained in the additional tab **Logic output**, which has the same appearance as shown in Fig.11.c.

4. The information contained in the **USS Settings** tab is displayed only in the SERVICE and ADJUSTMENT operating modes and is only available for editing in the ADJUSTMENT mode. In WORK mode, the tab is not available for viewing.
5. You can view the current values of the measured parameters on the **Flow Data** tab (Fig.12) after pressing the **Start reading** button.

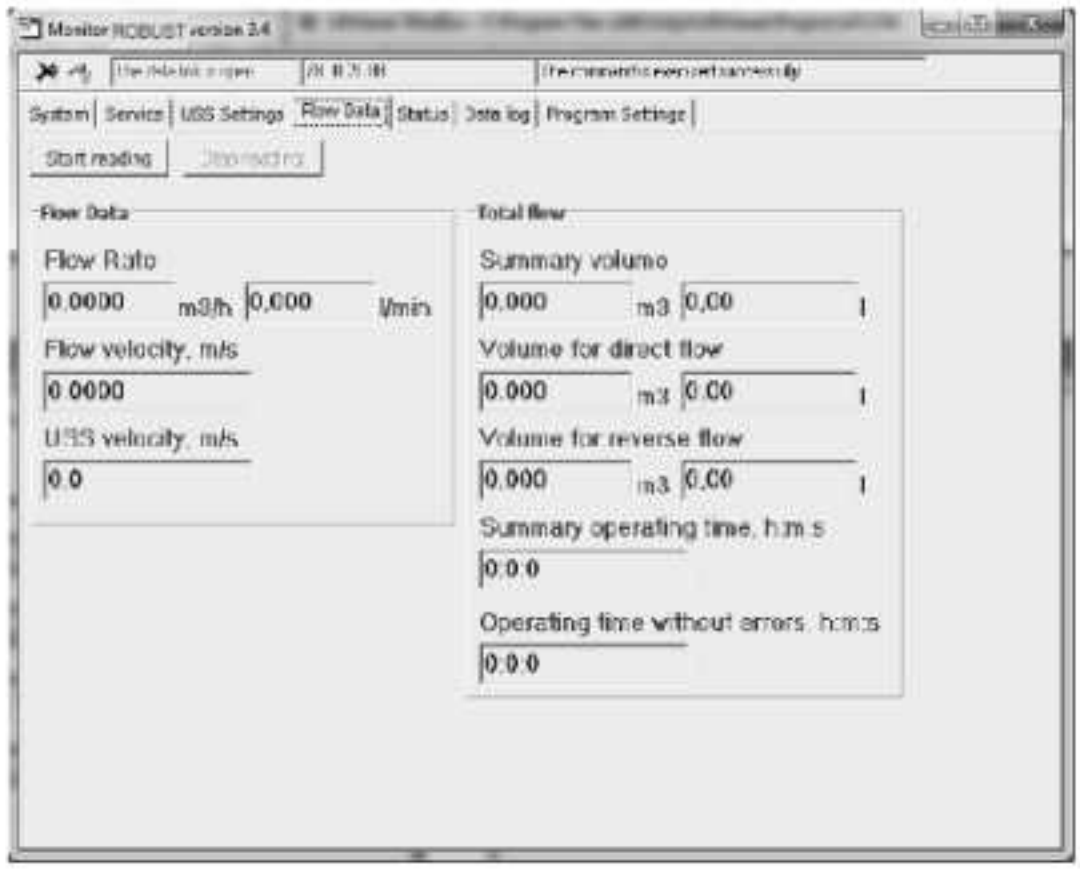



**Fig.12. View of the Flow Data tab**

4.1.5. The current state of the flow meter (presence/absence of emergencies) is displayed on the **Status** tab after pressing the **Start reading** button (Fig.13).

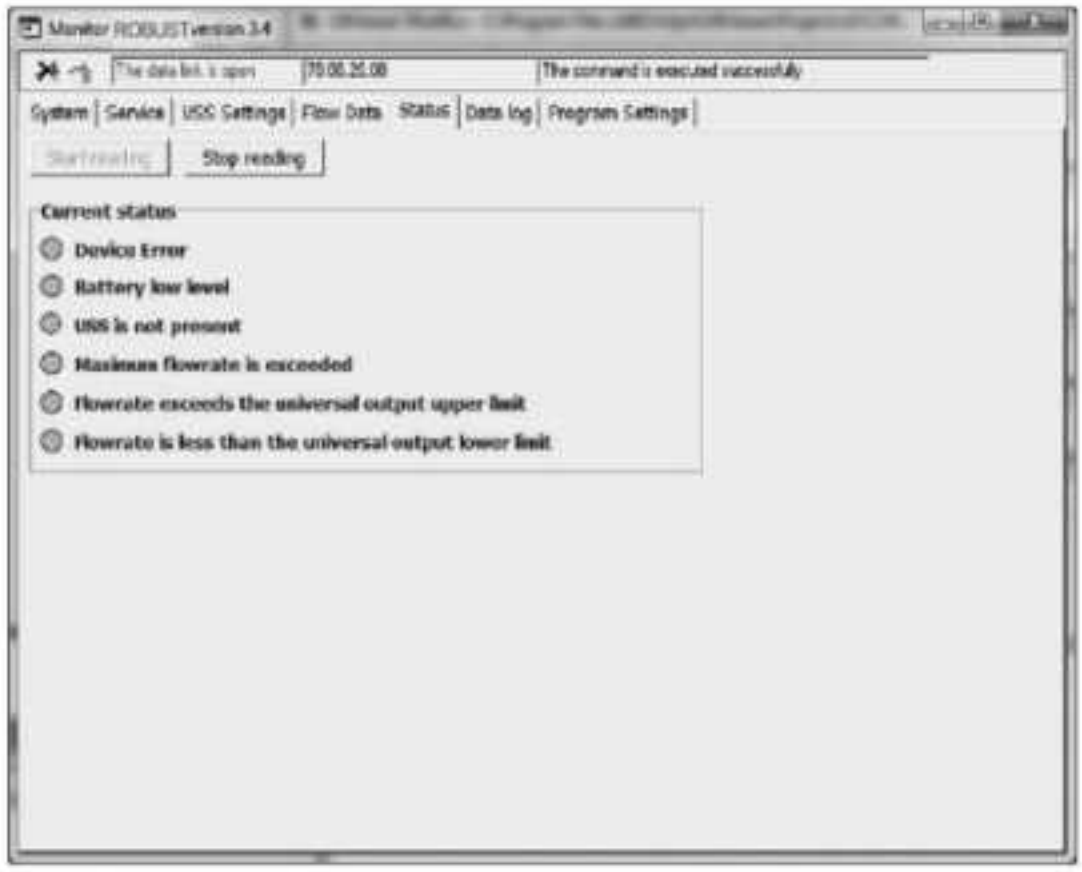



**Fig.13. View of the Status tab**

4.1.6. To read the archive data, go to the **Data log** tab (Fig.14), select the type of archive (hourly, daily or monthly), set the required interval in the **Date from** and **Time from** fields and press the **Read** button.

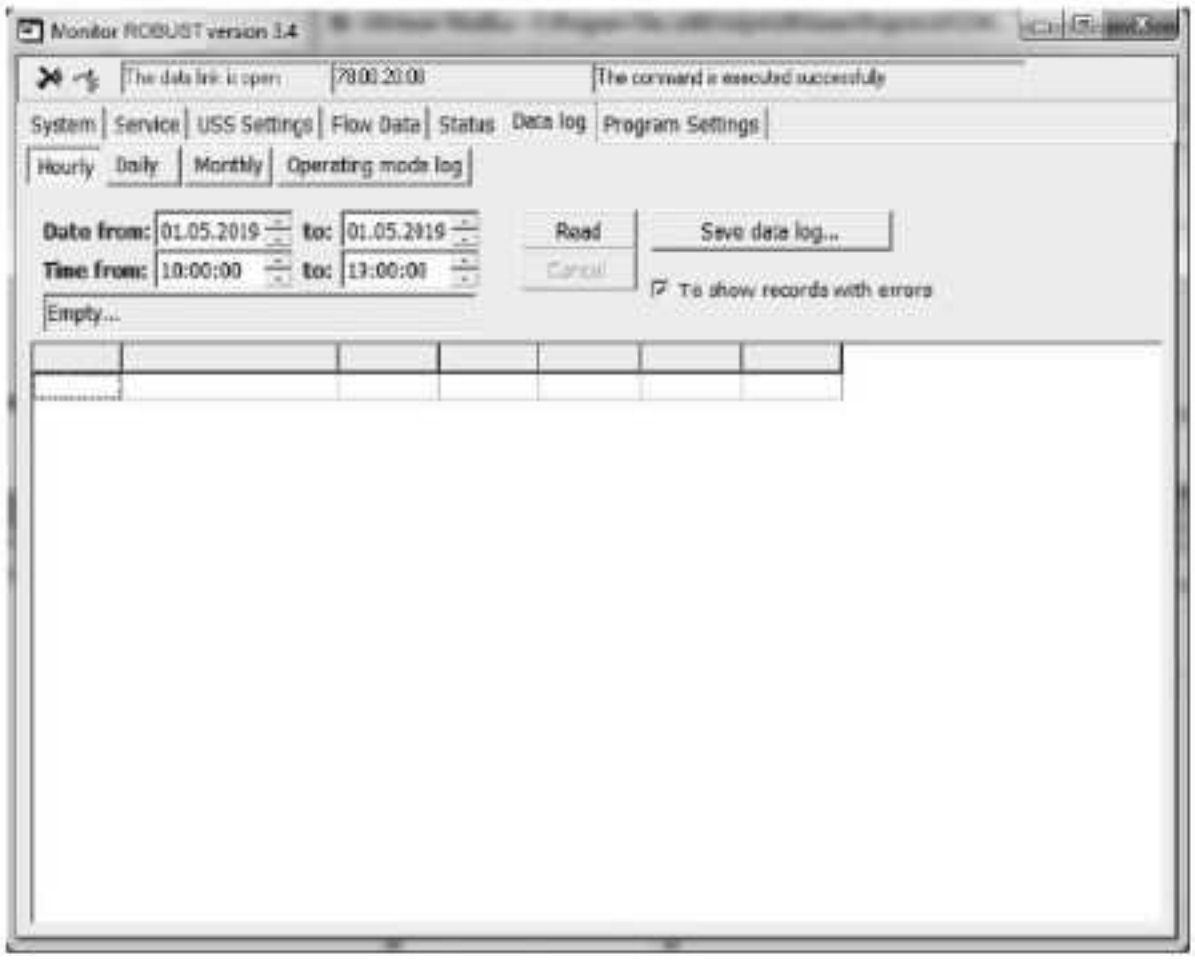



**Fig.14. View of the Data log tab**

7. Once the flow meter has been set up:

   - close the device communication channel with PC by pressing the in the upper left corner of the Monitor ROBUST application window;
   - close the Monitor ROBUST application window;
   - set the device to WORK mode: remove the jumper from the contact pair for allowing the modification of service parameters. Seal the contact pair.

## 2. Pre-commissioning

1. Pre-commissioning shall be performed by representatives of the organization having the right to perform such works or by representatives of the manufacturer.
2. After the installation of the flow meter, the pipeline is filled with the measured liquid. The flow meter's battery is connected, the flow meter is connected to the computer and adjusted in accordance with the instructions in section 4.1 of this manual.
3. The flow meter is ready for operation on initial start-up or after a long interruption in operation after:
   - A 30-minute flushing of the CP with fluid stream
   - A 30-minute warm-up of the flow meter.
4. When commissioning the flow meter, check:
   - The correct connection of the flow meter and the interacting equipment according to the selected connection diagram
   - The correctness of the set operating modes of the flow meter's universal and logic outputs.

   After commissioning, the SMC enclosure can be sealed to protect against unauthorized access during operation.

## 4.3. Operating procedure

The flow meter, which has been put into operation, operates continuously in automatic mode. Current values of measured parameters can be read from the flow meter LCD or via RS-485, M-Bus, Wireless M-Bus or NFC interfaces.

# 5. READING DATA WITH A SMARTPHONE

## 1. Hardware requirements

The ultrasonic flow meter of URSV- ROBUST version provides the possi- bility of connecting a smartphone based on Android OS of version 4.4 and higher with the support of NFC-V (ISO 15693).

For a list of NFC-enabled phones, see: https://devices.nfc.help/

Communication between the flow meter and smartphone is provided via the IOTAFLOW ROBUST monitor application.

## 2. Installing the IOTAFLOW ROBUST monitor application

The installation file for the IOTAFLOW ROBUSTmonitor application (apk.) is downloaded from vzljot.ru.

To install the application in Android 7+ versions, provide permission to the file manager as a source of software from an unknown source (see

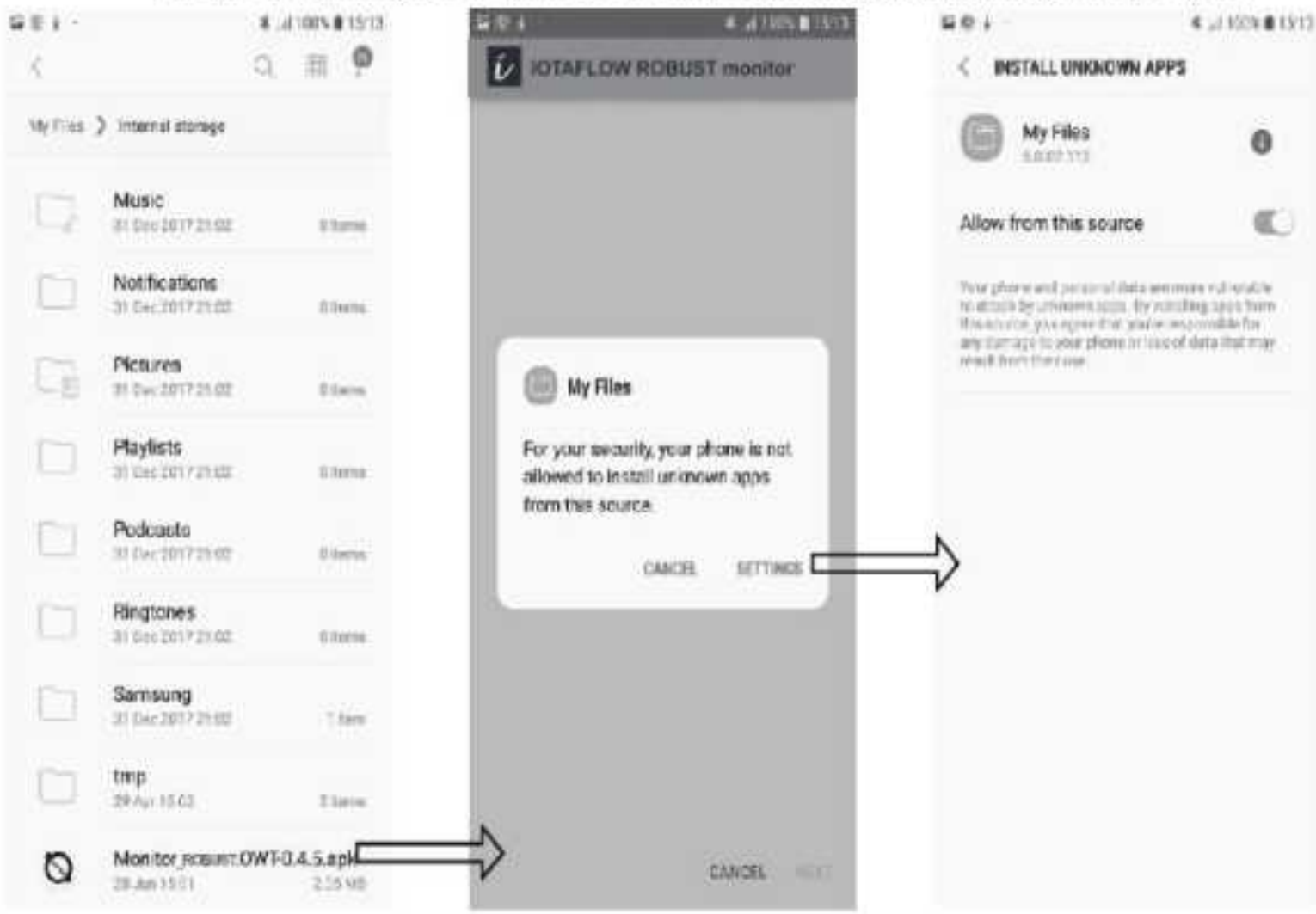



**Fig.15. Granting permission to the file manager**

Install the downloaded installation file using any file manager on your smartphone.

During the software installing, follow the hints of the system mes- sages that appear on the screen of your smartphone (Fig.16).

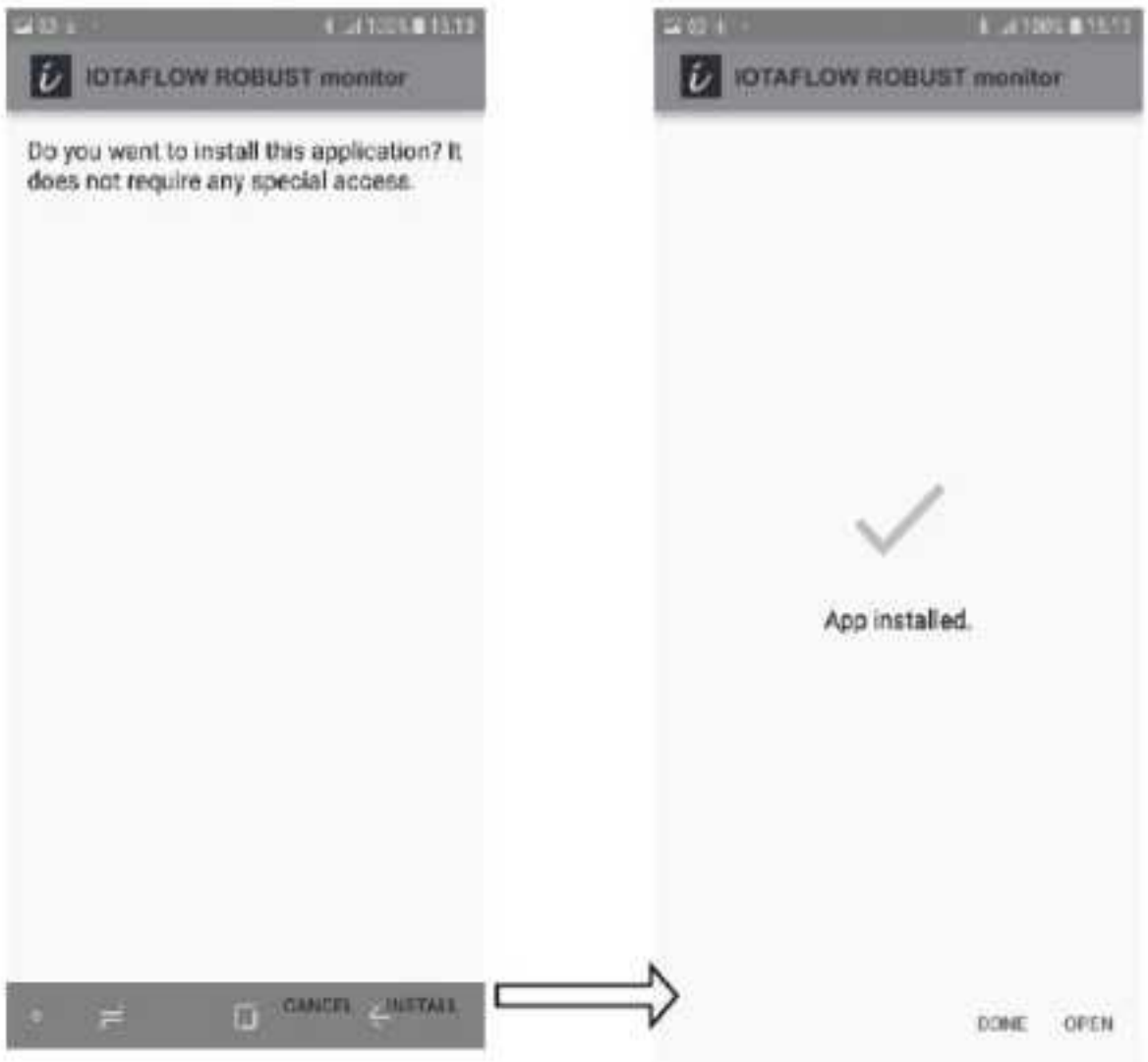



**Fig.16. IOTAFLOW ROBUST monitor application installation window**

When the installation process is complete, the IOTAFLOW ROBUST monitor software icon (Fig.17) should appear on the desktop and/or in the application menu.

**Fig.17. IOTAFLOW ROBUST monitor application icon**

## 5.3. Connecting to the flow meter

To use the **IOTAFLOW ROBUST** monitor program, enable the NFC function in the corresponding menu item (Fig.18) in the smartphone settings.

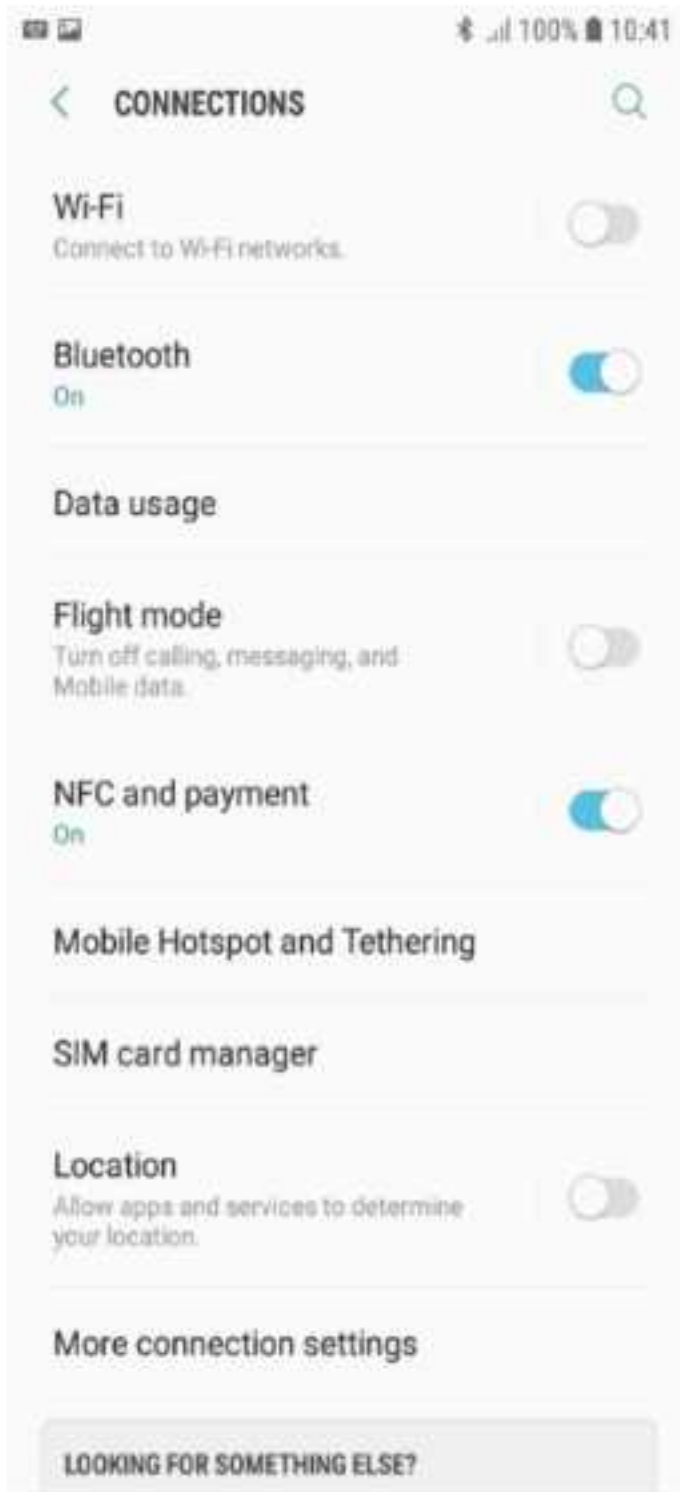



**Fig.18. Enabling NFC**

After installing the **IOTAFLOW ROBUST** monitor application, the application will request access to the file system of the smart phoneat its first start (Fig.19). Otherwise, saving of archives and its further use will be inacces- sible.

**Fig.19. Request for access to the file system**

## 5.4. Reading parameters

The application contains four tabs: **ABOUT UNIT**, **MEASUREMENTS**, **ARCHIVES** and **SAVED DATA** that can be swap sequentially on your smartphone screen by swiping left or right.

When you start reading the parameters by pressing the **READ** button, the progress bar in the form of a rotating circle will be displayed at the top of the screen (Fig.20).

**Fig.20. Display of data exchange**

A green circle indicates that the data is currently being read or a connection is being maintained.

Yellow circle (Fig.21) – the connection has been established, but now the smartphone is out of the NFC range.

**Fig.21. Display of the lack of NFC**

## 5.5. ABOUT UNIT tab

The appearance of the **ABOUT UNIT** tab is shown on Fig.22. You can read the parameter values by pressing the **READ** button.

**Fig.22. ABOUT UNIT tab window**

This tab displays the following parameters: software version of the flow meter, serial number of the flow meter, nominal diameter (DN), operating mode, checksum (CS) of the software, calibration factors, information about the use of universal and logic outputs, values of output factors (if the universal output works in the frequency or pulse mode).

## 5.6. MEASUREMENTS tab

The appearance of the **MEASUREMENTS** tab is shown on Fig.23.

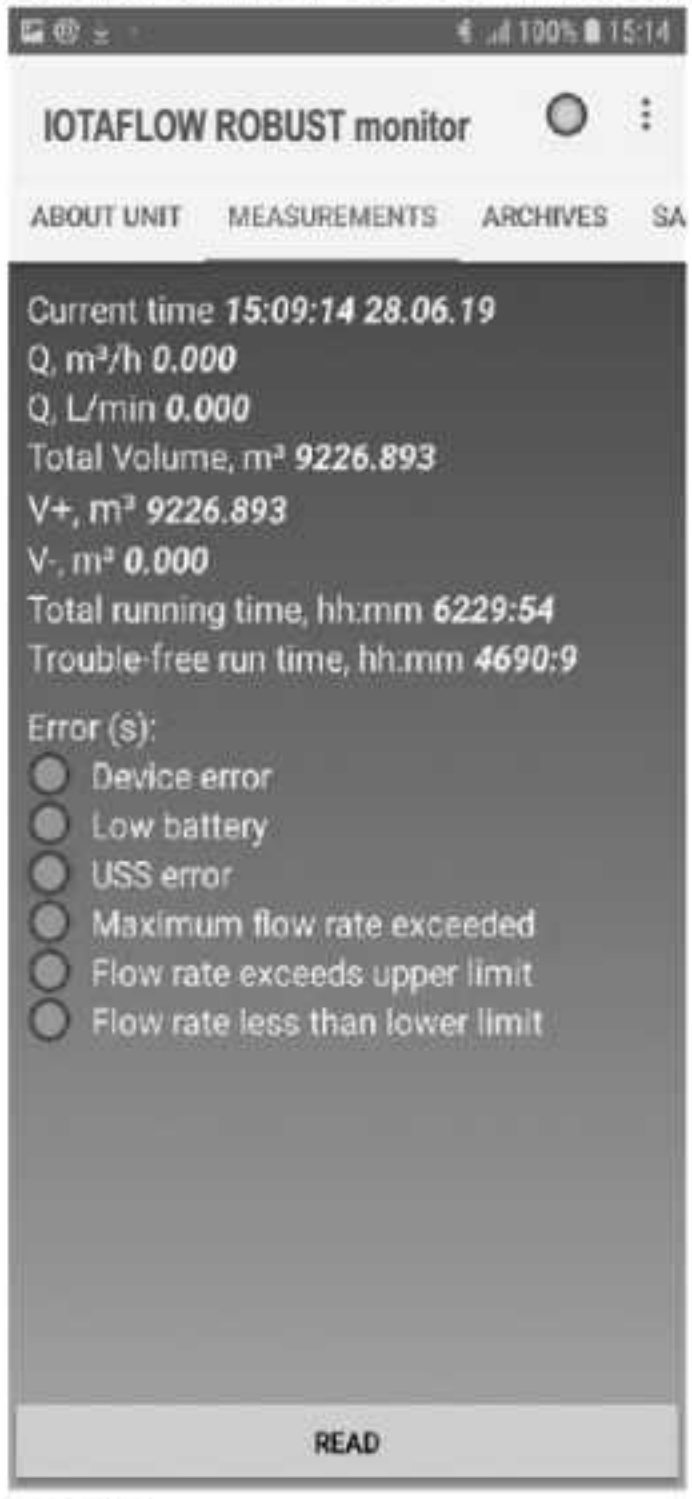



**Fig.23. MEASUREMENTS tab**

You can read current values by pressing the **READ** button.

The tab displays: the current date and time, the current flow rate value (in $m^3/h$ and l/min), accumulated volumes (direct, reverse and total), total operating time and trouble-free operation time, emergencies.

The total volume is calculated based on the obtained data on V+ and V-.

If there are no errors, all circles to the left of the name of the diagnosed ER are grayed out, and if an error occurs, the corresponding circle changes to red (e.g. Fig.24).

**Fig.24. Display of the emergency USS error**

## 5.7. ARCHIVES tab

The monitor allows you to read hourly (last 96 records), daily (last 92 records) and monthly (last 48 records) archives from the flow meter.

The appearance of the **ARCHIVES** tab is shown on Fig.25.

**Fig.25. ARCHIVES tab**

The tab is used to select the type of archive: hourly, daily or monthly. After that you should press the **READ** button. Thus, the window with records of the chosen archive (Fig.26) opens:

**Fig.26. Tab with records of the selected archive**

To view a certain record, click on the line with the corresponding date.

To save all records in the file, press the **SAVE** button. You will see a request to confirm that you want to save SMCV files (Fig.27):

**Fig.27. Confirming the file saving**

Saving the files is accompanied by the message "**SMCV file was created**" (Fig.28).

**Fig.28. Confirming the file saving**

The files are saved to: sdcard \ **IOTAFLOW** \ monitor \ export.

Depending on the type of smartphone, the path to the saved archive files may differ.

NOTE. The file name is formed with indication of the archive type and time of reading the record from the device (Fig.29):

**Fig.29. Saved archive files**

A message is displayed on the smartphone screen (Fig.30) when you are finished with the application:

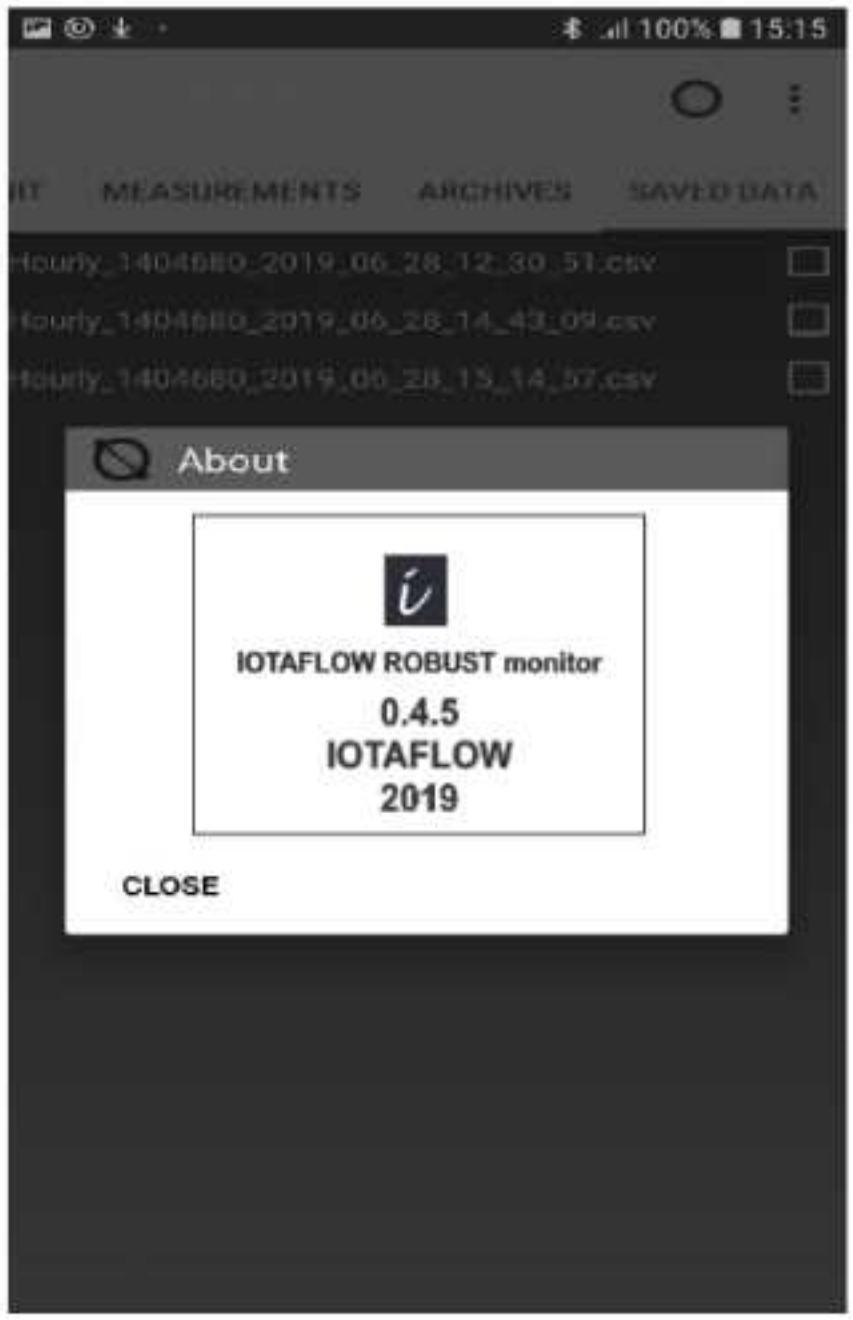



**Fig.30. Closing the application**

# 6. MAINTENANCE

## 1. Technical condition monitoring

1. Commissioned flow meter is recommended to be periodically inspected to check:
   - Flow meter performance
   - Compliance with operating conditions
   - Availability of supply voltage
   - Absence of external damage to the flow meter
   - Reliability of electrical and mechanical connections.

   Inspection frequency depends on operating conditions but should not be less than once a month.

   At least once a year, the internal CP channel should be inspected for contamination and/or deposits. A light reddish patina is allowed, which should be removed with a clean soft cloth soaked in water during the maintenance.

   If there are any other types of dirt or deposits or if their thickness is significant, the CP surface should be cleaned with water, clean cloths and non-abrasive detergents immediately after removing the flow meter from the pipeline.

   The significant contamination on the CP surface in contact with the liquid indicates the unsatisfactory condition of the pipeline.
2. Failure to comply with the operating conditions of the flow meter specified in clauses 1.2.5 and 2.1, external damage or damage to the internal CP surface may result in failure of the flow meter or exceeding the permissible level of measurement error.

   In case of damage of the product or communication cable it is necessary to contact the manufacturer to determine the possibility of its further operation.
3. The performance of the device is determined by the presence and content of the indication on the flow meter display or on the PC monitor. Possible malfunctions diagnosed by the flow meter are specified in clause 6.2.
4. Taking into account the operating conditions and the version, the flow meter refers to the products that are repaired by the manufacturer.
5. The flow meter must be sent for verification or repair with the device passport. The accompanying documents shall contain the sender's postal details, telephone and fax number, as well as the method and address of return delivery.

   When sending the device for calibration or repair, clean the internal CP channel from deposits, scale, as well as from the remnants of working fluid after dismantling.

## 2. Troubleshooting

1. List of possible faults, failures and alarm situations detected by the flow meter and displayedon the LCD with "**X**" character in the "**ERR = - - - - - -**" line is given in Table 7. A sign-position code is read from ***right*** to ***left***.

**Table 7**

| Sign position number | Explanation |
|---|---|
| 1 | Devise Error |
| 2 | Battery low level |
| 3 | USS is not present |
| 4 | Maximum flowrate is exceeded |
| 5 | Flowrate exceeds the universal output upper limit |
| 6 | Flowrate exceeds the universal output lower limit |

2. The same information can be read via the communication interface from the **Status** tab of the "Monitor ROBUST" program (see section 4.1.5). In the event of a fault, the LED on the corresponding line lights up red.

3. If **X** symbol is displayed in position 1 or the indicator lights up in red in the **Devise Error** line in the **Status** tab, the flow meter shall be forwarded for repair.

4. If **X** is displayed in other positions, or the lights are red in other lines of the **Status** tab, and/or in case of no pulses at Universal output, check the following:

- Liquid is present and running through the pipeline
- Air is not collected in the flow meter's location;
- The universal output conversion factor and flow rate cutoffs are correct; change their values if necessary.

If the above checks are positive, you should contact the device manufacturer to determine the possibility of its further operation.

6.2.5. Past emergencies are recorded in the device archive and can be viewed via the serial interface in the **Data log** tab, selecting the type of archive and time interval. The assignment of the 16-position code in the column of emergency situations of the archive from the 1st to 6th positions (from right to left) corresponds to the one specified in Table 7. The presence of a fault is indicated by the number "1", while the absence is indicated by the number "0".

# 7. PACKING, STORAGE AND TRANSPORTATION

1. The flow meter is packed in a separate container (corrugated carton or wooden box).

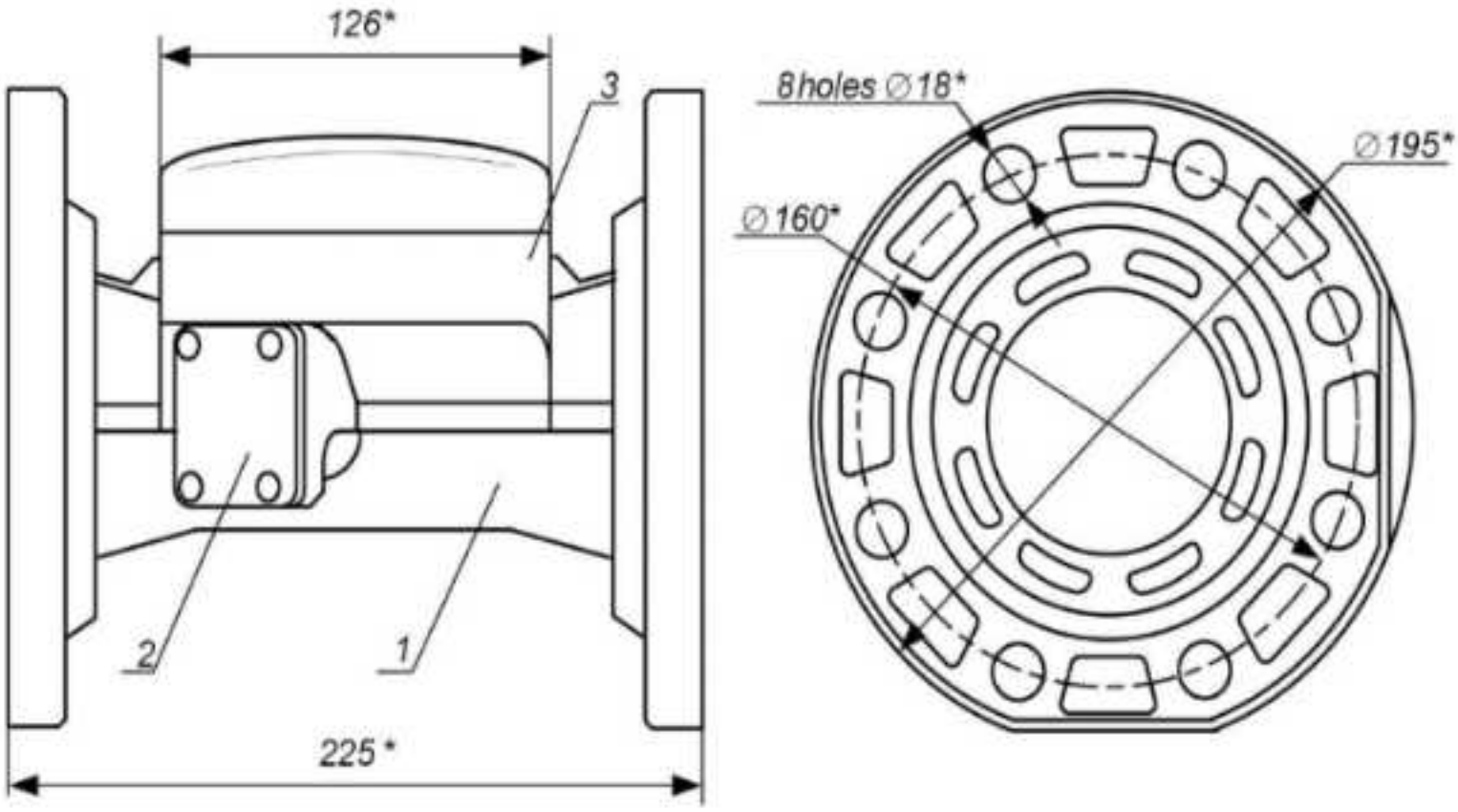
2. The flow meter should be kept in the manufacturer's box in a dry heated storeroom. The storeroom should be free from current-conductive dust, acid or alkali fumes and aggressive gases.

   During storage the flow meter does not require any special maintenance.

3. The flow meters can be transported by road, rail, sea or air (except for unsealed cargo compartments) provided that the following requirements are met:

- Flow meter is transported packed in the manufacturer's box
- Protection against moisture is provided
- Temperature is within the range of -25 to +50 C
- Humidity does not exceed 98 % at +35 C
- Vibration is within the range of 10-500 Hz with maximum 0.35 mm amplitude or 49 $m/s^2$
- Impact acceleration does not exceed 98 $m/s^2$
- Flow meters are fixed to prevent damages.

# APPENDIX A. Appearance of the flow meter



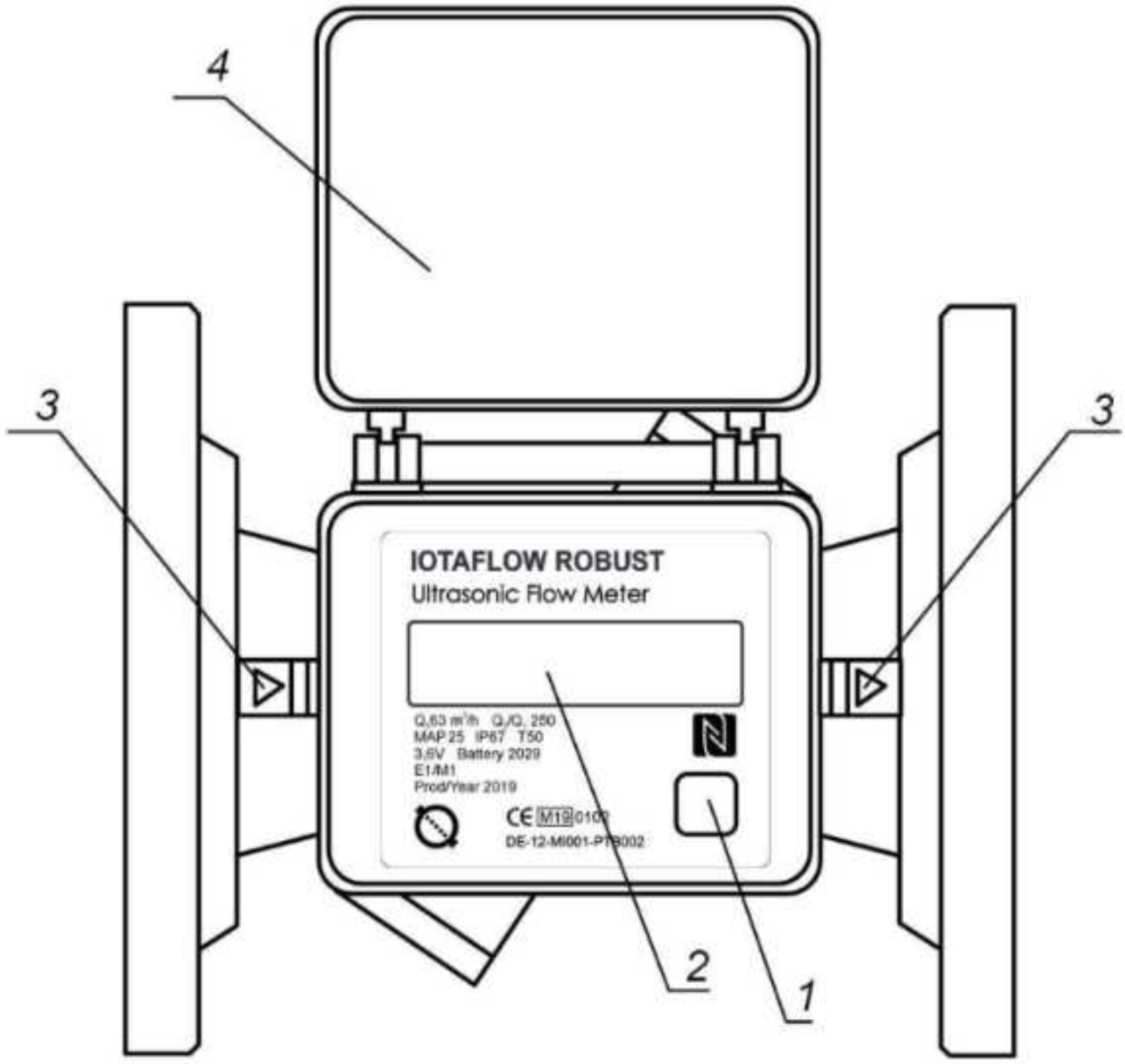

a) front view

b) side view

* Reference size

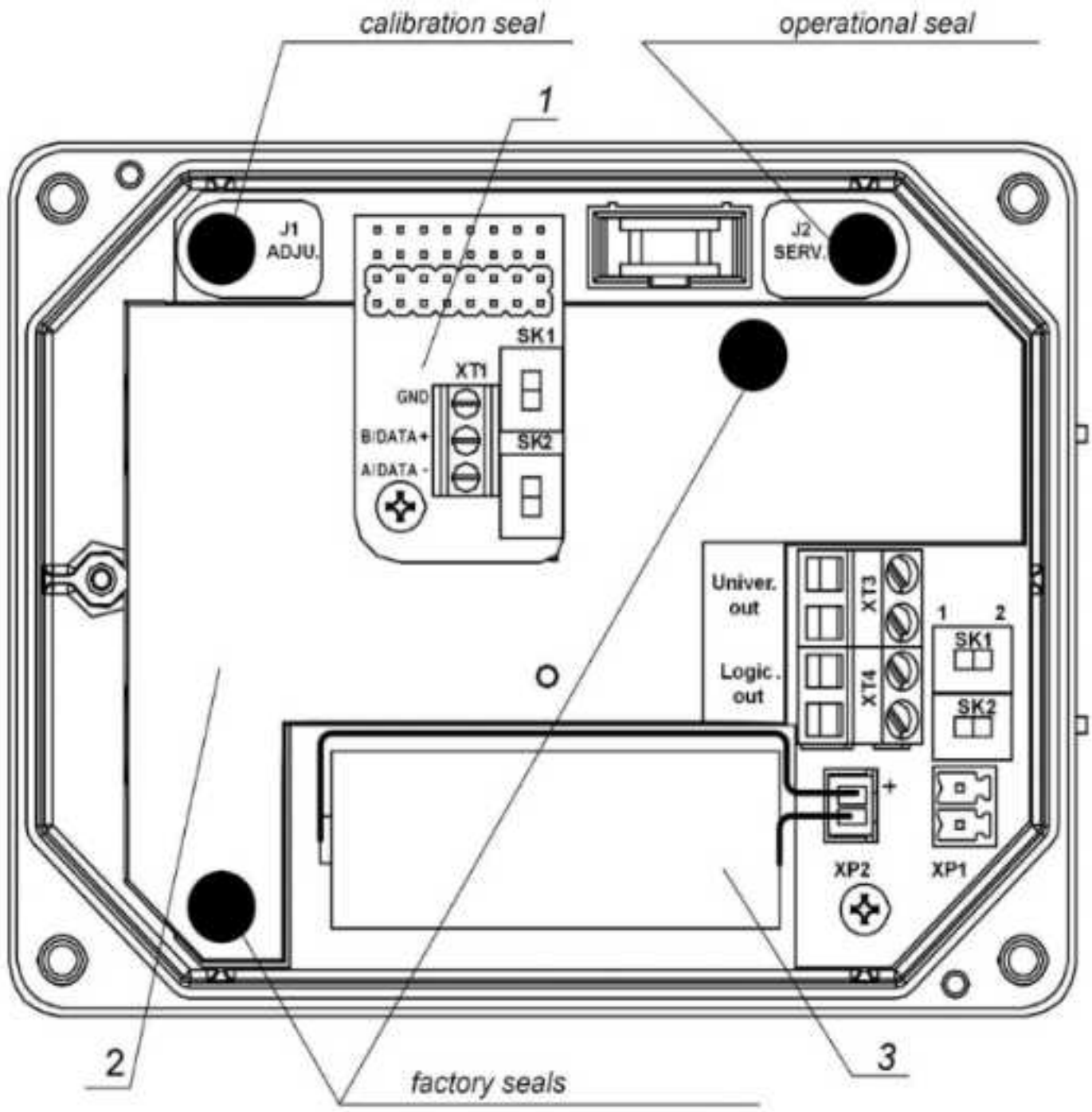
*1 – pipeline section; 2 – PEA; 3 – secondary converter.*

**Fig.A.1. Flow meter IOTAFLOW ROBUST of U-ROBUST version.**




*1 – button; 2 – LCD; 3 – flow direction indicator; 4 – cover.*

**Fig.A.2. Flow meter IOTAFLOW ROBUST . Front panel view**




*1 – RS-485 interface module; 2 – display; 3 – built-in 3.6 V battery;*

XT1 – RS-485 interface output connector;

SK1, SK2 – flow meter network address setting switches (on the RS-485 interface module);

XP2 – 3.6 V battery connector;

XT3 – universal output connector;

XT4 – logic output connector;

J1, J2 – contact pairs of the flow meter operation mode;

SK1, SK2 – switches for setting the operating modes of the universal and logic outputs.

**Fig.A.3. View of the secondary converter processing module with RS-485 interface module**

# APPENDIX B. Wiring diagrams

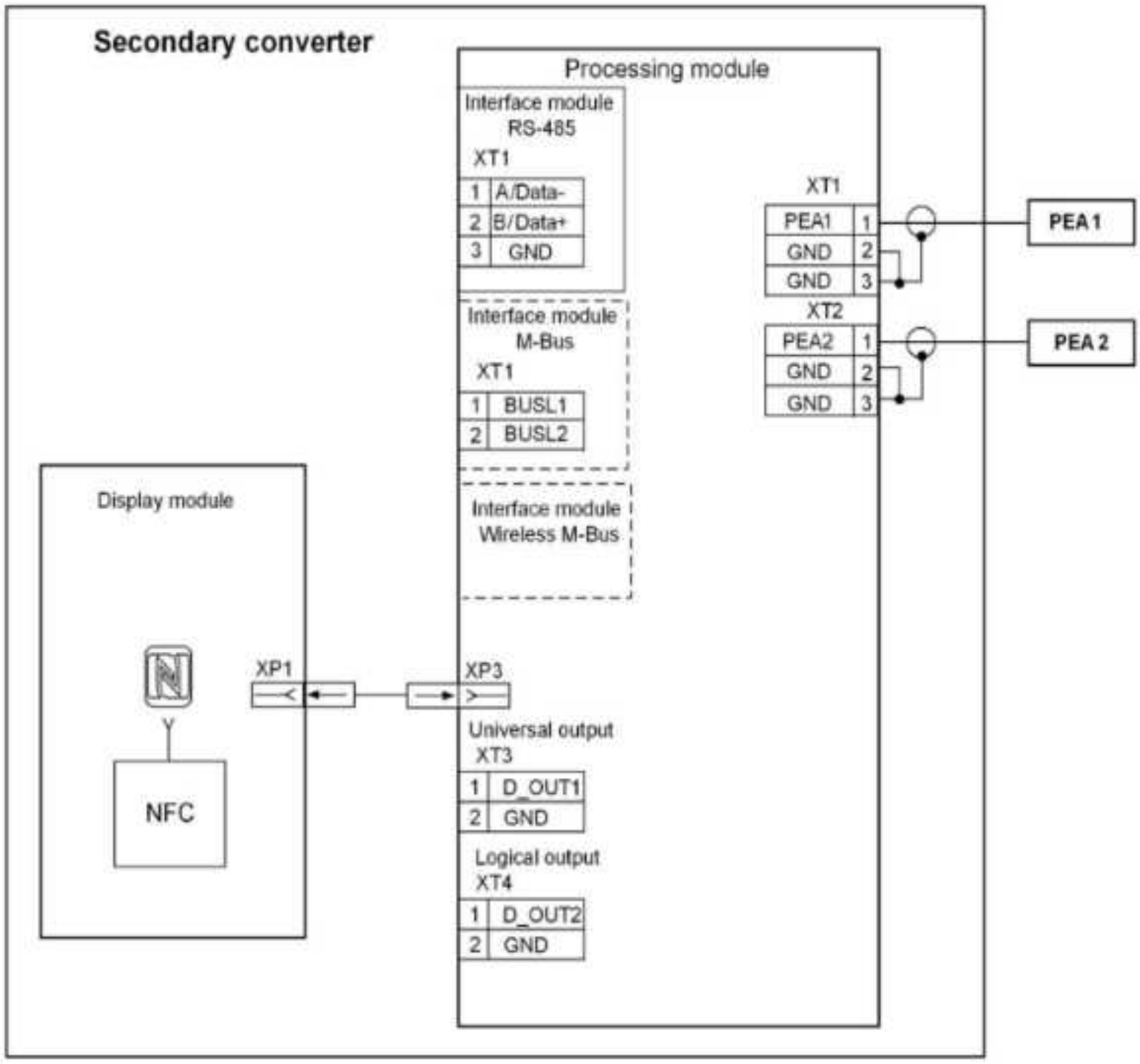



**Fig.B.1. IOTAFLOW ROBUST flow meter wiring diagram**

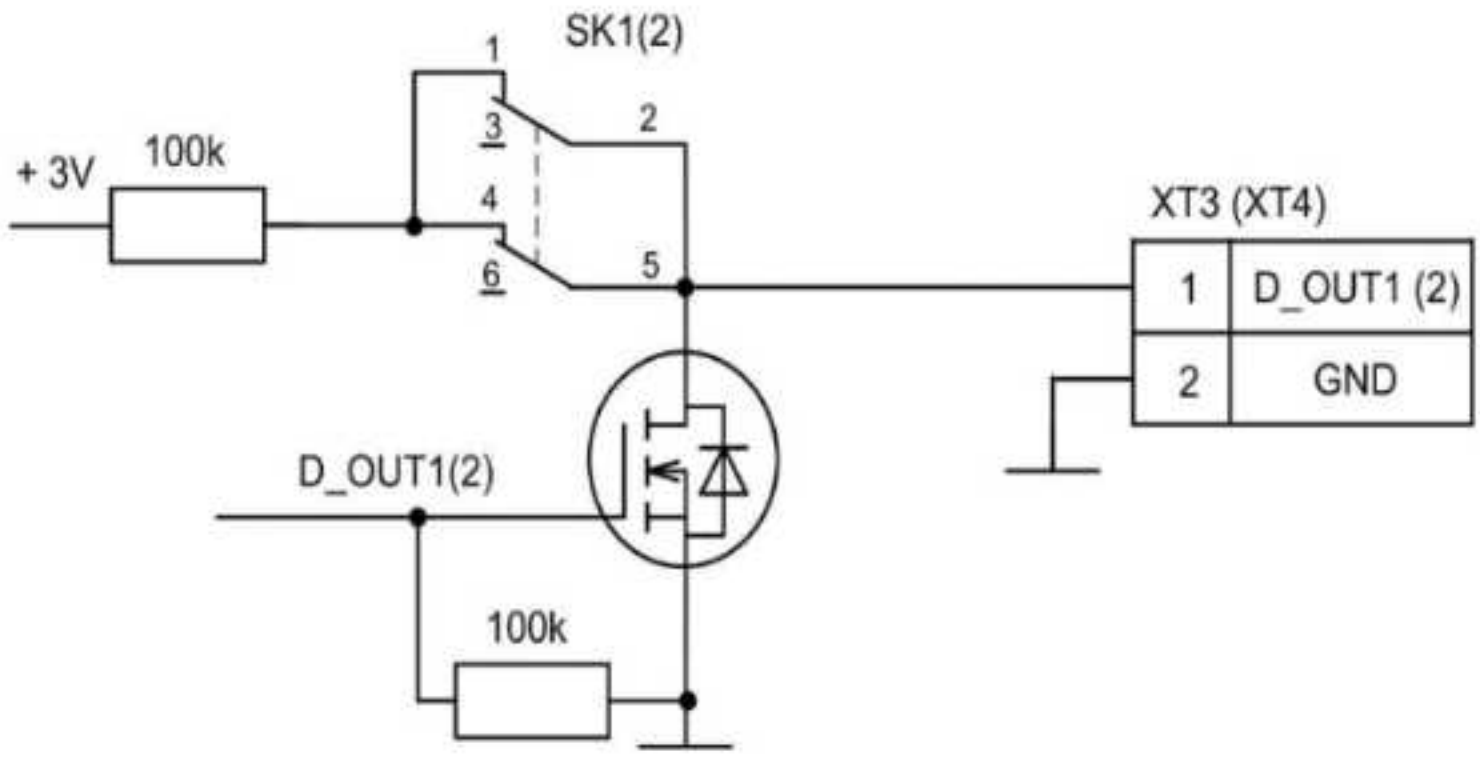



**Fig.B.2. Diagram of the terminal stage of the universal and logical outputs**

The terminal stage of the universal and logic outputs (Fig.B.2) can be powered either from the internal power supply (active mode) or from an external power supply (passive mode). Typical delivery includes passive operation of the terminal stage.

In the active mode and at the set value of the parameter **Active level – Hihg**, the output voltage in the logic mode and the pulse amplitude in the frequency (pulse) mode, depending on the resistance of the load, can be from 1.7 to 3.6 V. If there is no pulse, and the **Low** level is in the logic mode, the output voltage is not more than 0.4 V. Operation of the output in active mode is allowed for a load with a resistance of at least 200 kOhm

The terminal stage is connected to the internal +3.6 V power supply via the SK1 (SK2) switch.

The length of the communication line for the universal and logical outputs is up to 300 m.

# APPENDIX C. Parameters available for editing in Monitor ROBUST application

**Table C.1. System tab (Fig.9)**

| Parameter designation on the PC monitor | Parameter, measurement units | Possible values | Value after initialization |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| **Address** | Address of the flow meter in the network | **1-247** | **1** |
| **Boudrate** | Baud rate in the network of devices | **1200, 2400, 4800** | **4800** |
| **Time and Date** | Current device time and date | **XX/XX/XX** (dd.mm.yy) **XX:XX:XX** (hh:mm:ss) | No changes available |
| **Switching mode** | Clock transition between "Summer" and "Winter" time | **no switch standard user** | **standard** |
| **Data log clear** | Reset of accumulated archives | **...; start** | **...** |
| **Total flow reset** | Zeroing of accumulated volume meters | **...; start** | **...** |

**Table C.2. Service tab (Fig.10)**

| | | | |
|---|---|---|---|
| **Median mean** | Number of measurements to determine the median value | **1-21** | **7** |
| **Arithmetic mean** | Number of measurements to determine the arithmetic mean | **1-50** | **25** |
| **Inertion time, s** | Minimum duration of USS loss not fixed in the archives | **0-255** | **10** |
| **Cutoff flowrate, $m^3/h$** | Minimal flow rate cutoff | **0.000-999.000** | **0.250** |
| **Fast reaction** | Dynamic flow measurement mode | **On/Off** | **Off** |
| **Threshold, %Qmax** | Criterion for quick installation | **0-100** | **5** |
| **Contrast** | Adjusting the indicator contrast | **1-63** | **28** |
| **Updating time, s** | Measured information update period on the indicator | **1-30** | **10** |
| **Worktime, s** | Time after which the indicator automatically turns off after the last pressing of the button | **1-255** | **30** |
| **Dimention** | Flow rate dimension | **$m^3/h$; l/min** | **$m^3/h$** |

**Table C.3. Service / Universal output tab (Fig.11)**

| Parameter designation on the PC monitor | Parameter, measurement units | Possible values | Value after initialization |
|---|---|---|---|
| **Type** | Universal output operation mode | **Off**<br>**Logic**<br>**Pulse**<br>**Frequency** | **Off** |
| | **Type – Frequency (Fig.11.a)** | | |
| **Flow** | Output assignment (parameter designation) | See Table C.4 | **any direction** |
| | **Parameters** tab | | |
| **Maximum output frequency, Hz** | Maximum output frequency | **0-10** | **10** |
| **Error frequency, Hz** | Output frequency at Q $Q_{max}$, | **0-10** | **0** |
| **Lower limit by florate, m³/h** | Lower flow rate threshold for universal output in frequency mode | **0-999.999** | Depending on DN |
| **Upper limit by florate, m³/h** | Upper flow rate threshold for universal output in frequency mode | **0-999.999** | Depending on DN |
| **Output conversion factor, p/m³** | Output conversion factor | **0,001- 200000.000** | Depending on DN |
| **Conversion factor calculation** | Output conversion factor calculation | **...; Start** | **...** |
| **Current frequency value, Hz** | Current frequency value | **0-10** | Corresponds to flowrate value |
| **Active level** | Output voltage level when a signal is present (logic unit) | **"Low"**<br>**"High"** | **"Low"** |
| | **Type – Pulse (Fig.11.b)** | | |
| **Volume** | Output assignment (parameter designation) | See Table C.4 | **any direction** |
| | **Parameters** tab | | |
| **Pulse weight m³/pulse** | Pulse weight | **0.000001-100.0000** | Depending on DN |
| **Pulse width, ms** | Pulse width | **1-1000** | **100** |
| **Up. limit by flowrate, m³/h** | Upper flow threshold for universal output in pulse mode, | **0-999.999** | Depending on DN |
| **Pulse weight calcul.** | Pulse weight calculation | **...; Start** | **...** |
| **Active level** | Output voltage level when a signal is present (logic unit) | **"Low"**<br>**"High"** | **"Low"** |

**Continuation of table C.3**

| Parameter designation on the PC monitor | Parameter, measurement units | Possible values | Value after initialization |
|---|---|---|---|
| **Type – Pulse (Fig.11.c)** | | | |
| **Output function** | Output assignment (parameter designation) | See Table C.4 | **Reversing flow direction** |
| **Active level** | Output voltage level when a signal is present (logic unit) | **"Low"**<br>**"High"** | **"Low"** |

NOTE. The purpose of the logic output is similar to that of the universal output in logic mode.

**Table C.4. Possible settings for universal output**

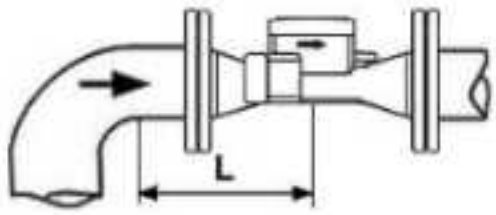
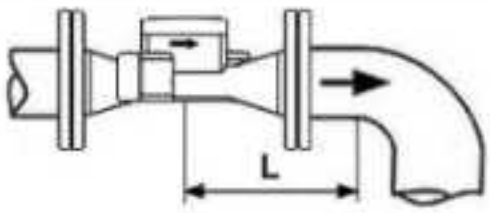
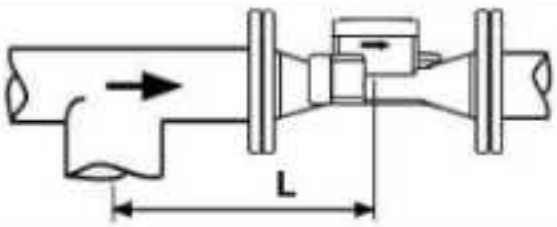
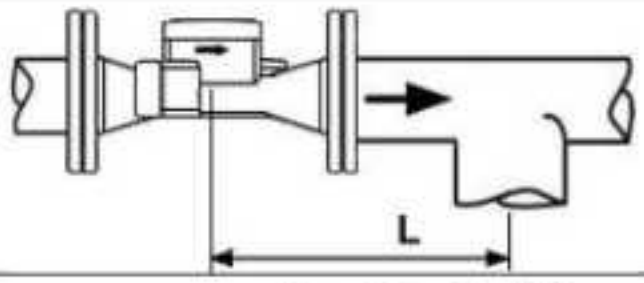
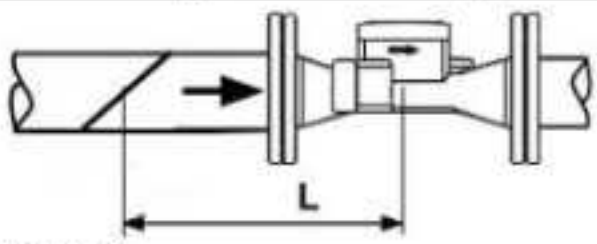
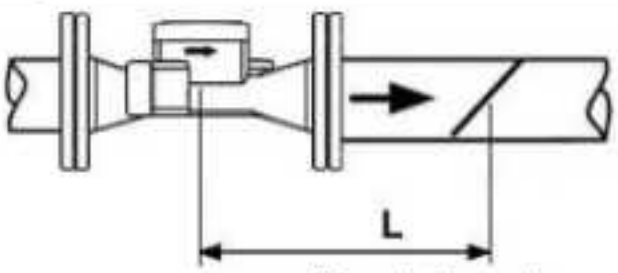
| Parameter | Possible output assignments when Type parameter is set | | | |
|---|---|---|---|---|
| | **off** | **frequency** | **pulse** | **logic** |
| **Off** | + | | | |
| **Flow - any direction** | | + | | |
| **Flow - direct flow** | | + | | |
| **Flow - reverse flow** | | + | | |
| **Volume - any direction** | | | + | |
| **Volume - direct flow** | | | + | |
| **Volume - reverse flow** | | | + | |
| **Reversing flow direction** | | | | + |
| **No ultrasonic signal** | | | | + |
| **Flowrate exceeds the hidh set-point** | | | | + |
| **Any Error** | | | | + |
| **Battery low level** | | | | + |

NOTE. The "+" sign shows one of the possible assignments (conditions) of the universal output.

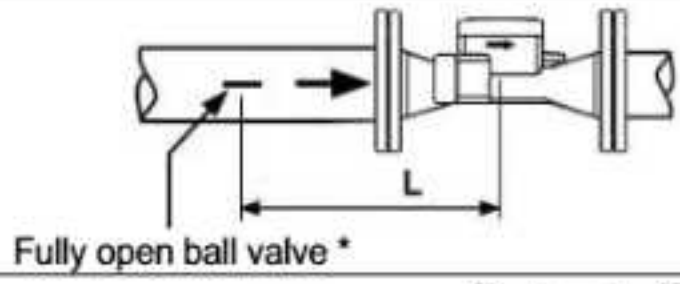
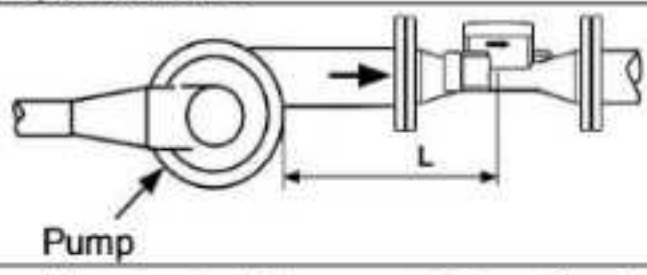
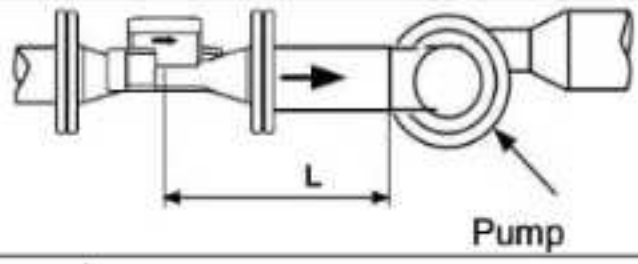
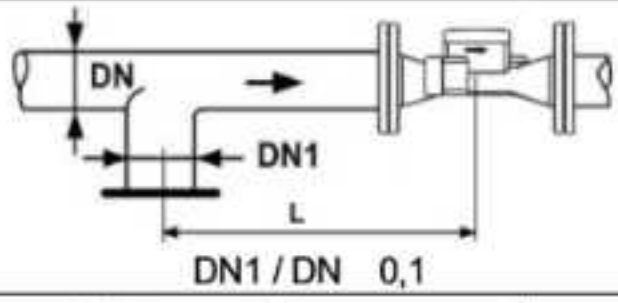
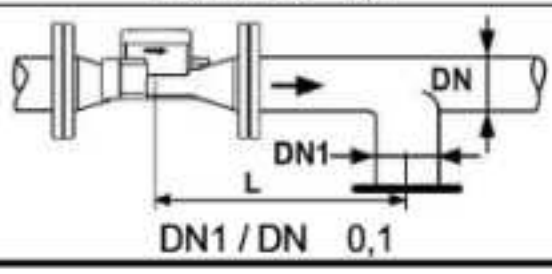
# APPENDIX D. Relative length of straight runs

Table D.1 shows the minimum values of the relative length of straight pipe sections for the various types of local hydraulic resistances required for installation of the flow meters.

**Table D.1**

| Type of local hydraulic resistance | Relative length of a straight run, L, minimum |
|---|---|
| 1 | 2 |
|   | 10·DN |
|   | 3·DN |
|   | 10·DN |
|   | 10·DN |
|   Regulating valve | 30·DN |
|   Regulating valve | 3·DN |

**Continuation of table D.1**

| 1 | 2 |
|---|---|
|  | 10·DN |
|  | 30·DN |
|  | 3·DN |
|  | 10·DN |
|  | 3·DN |

* - fully open, full port ball valve is not a hydraulic resistance.

If there are several hydraulic resistances in the pipeline, the length of the straight section of the pipeline to the nearest resistance to the PEA must be at least as specified in this table, and the distance from the PEA to each of the other hydraulic resistances must be at least as large as the value given in the table for this type of resistance.

NOTE. If there is no hydraulic resistance in the pipeline, the flow meter should be installed taking into account the lengths of straight sections 3·DN before the flow meter and 2·DN after the flow meter.

# APPENDIX E. Flow straightener design

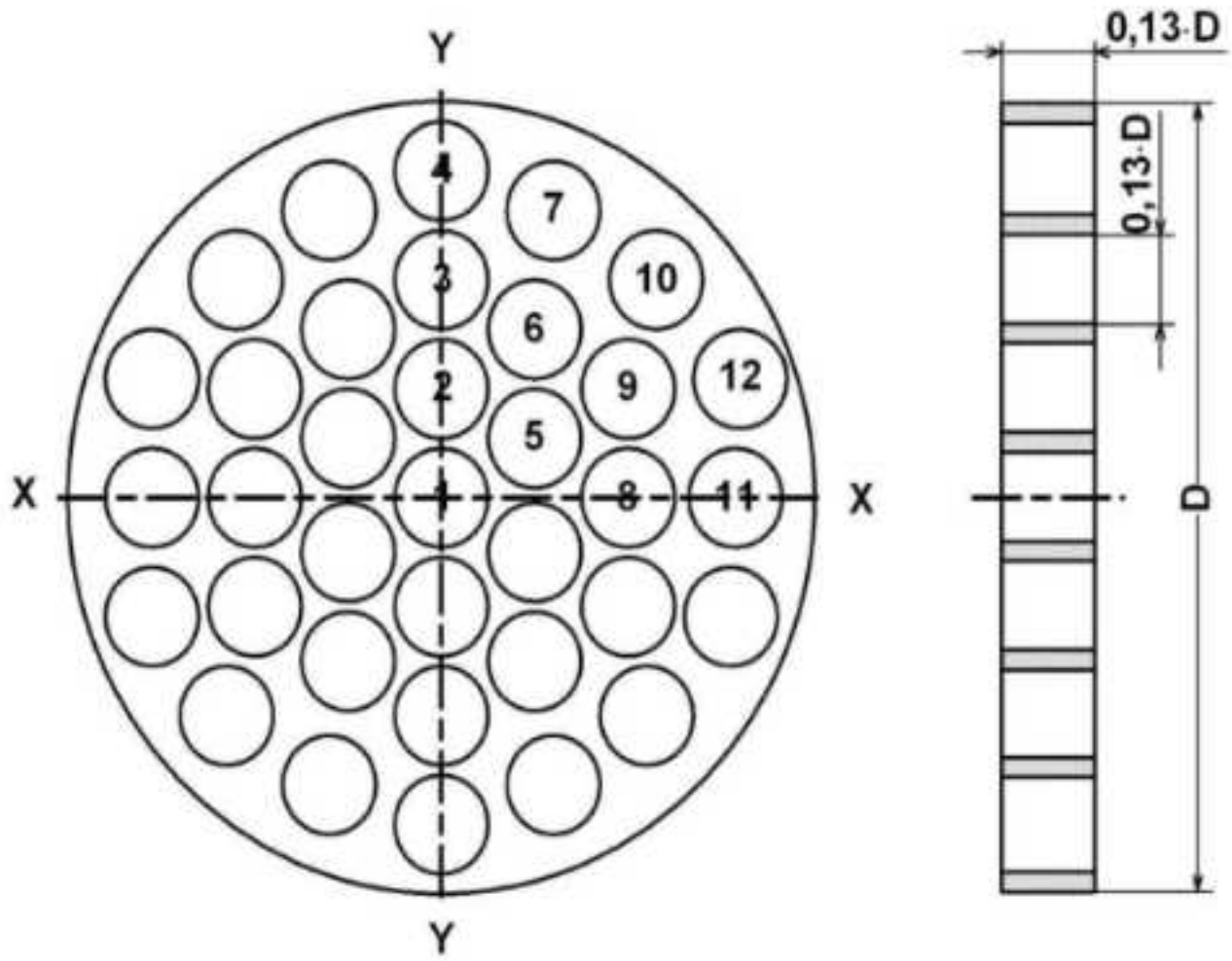
1. Fig.E.1 shows the design of flow straightener of A type that is manufactured according to the following rules:

a)the thickness of the flow straightener board is equal to the diameter of the holes; depending on the material, the board may consist of one or more plates;

b) all hole diameters in the board are the same;

c)holes are more densely distributed in the center of the board and rarer in the periphery;

d) holes on the flow inlet side are chamfered.




*D – internal diameter of the pipeline where the flow straightener is inserted.*

**Fig.E.1. Design of flow straightener of A type**

2. To reduce the weight and amount of material, the flow straightener of B type (Fig.E.2) can be used according to the following rules:

a) tubes are inserted into the board holes;

b) length of the tubes is equal to the diameter of the tubes;

c) all hole diameters in the board are the same;

d)holes are more densely distributed in the center of the board and rarer in the periphery;

e) holes on the flow inlet side are chamfered.

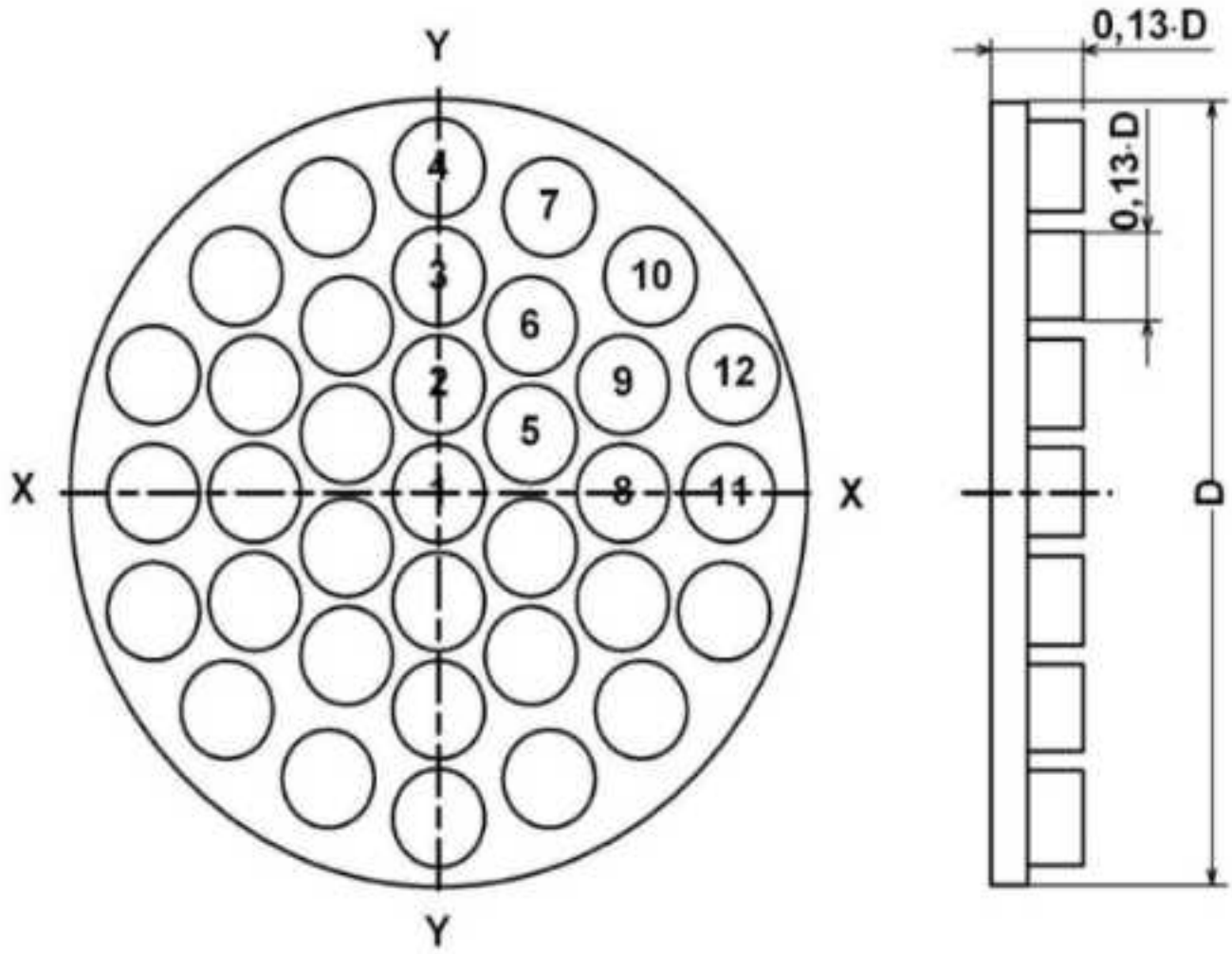



**Fig.E.2. Design of flow straightener of B type**

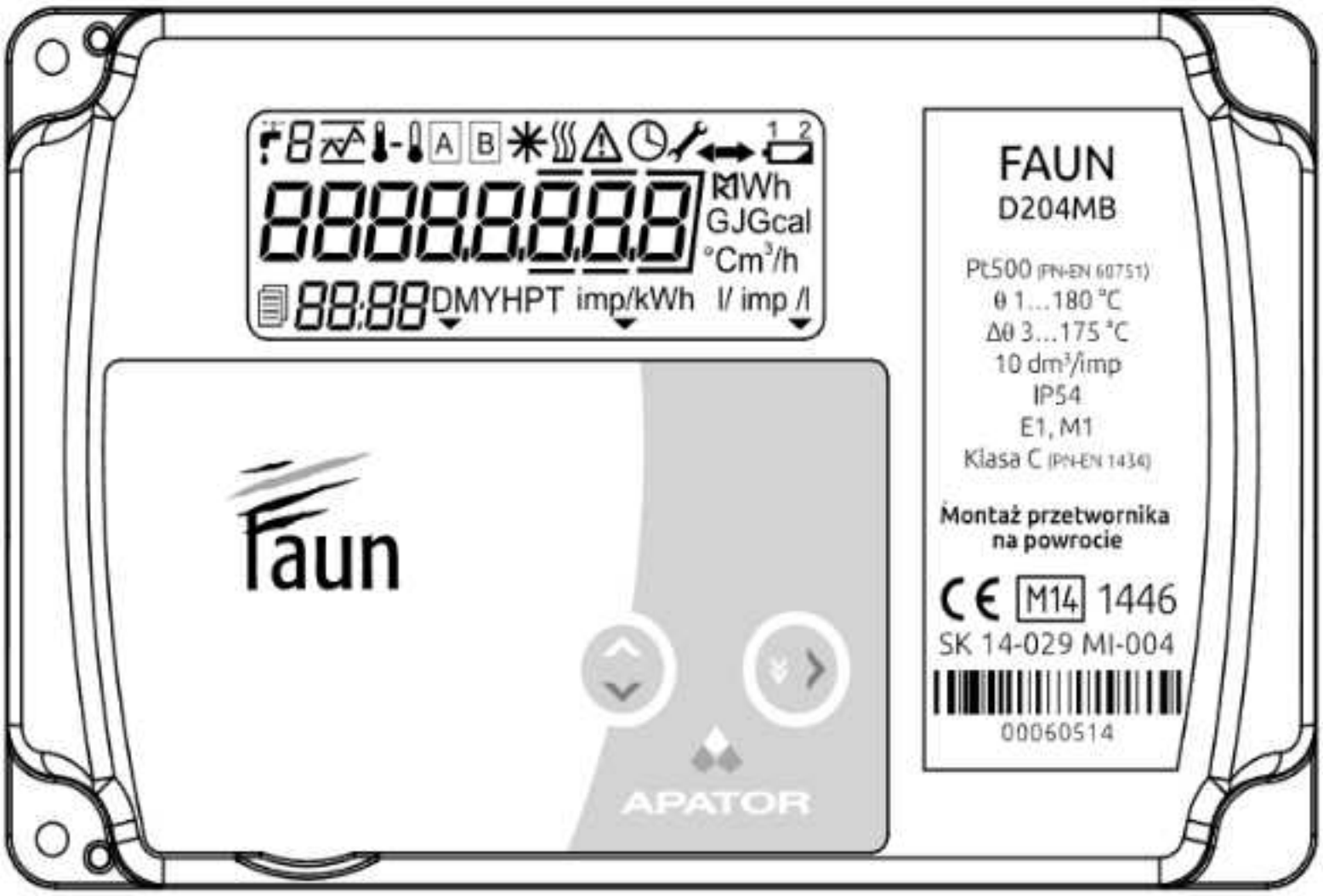
3. Hole pattern in the flow straighteners is shown in Table E.1.

**Table E.1. Hole coordinates in the flow straighteners of A and B type (D is pipe inner diameter)**

| No. | X-axis | Y-axis |
|---|---|---|
| 1 | 0 | 0 |
| 2 | 0 | 0,142 D |
| 3 | 0 | 0.283 D |
| 4 | 0 | 0.423 D |
| 5 | 0.129 D | 0.078 D |
| 6 | 0.134 D | 0.225 D |
| 7 | 0.156 D | 0.381 D |
| 8 | 0.252 D | 0 |
| 9 | 0.255 D | 0.146 D |
| 10 | 0.288 D | 0.288 D |
| 11 | 0.396 D | 0 |
| 12 | 0.400 D | 0.151 D |

4. The flow straightener is installed in the pipeline at 1 2 DN of the pipeline from the local resistance that is the last in the flow. When installing the flow straightener, the required length of the straight section in front of the PEA is defined as the distance from the hydraulic resistance to the PEA.

# FAUN




# Installation and Setup Instructions

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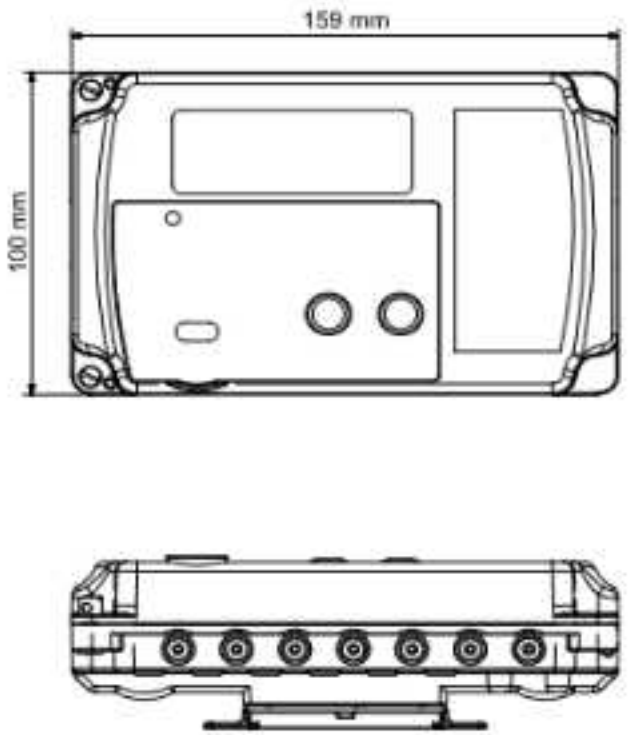


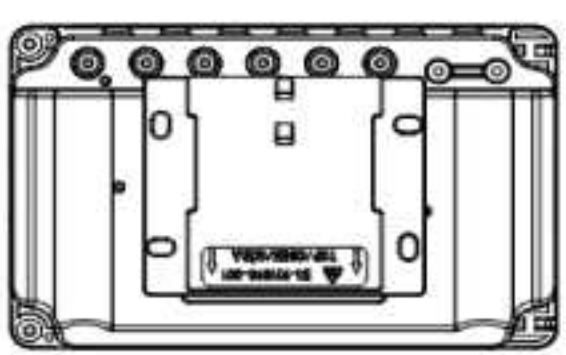

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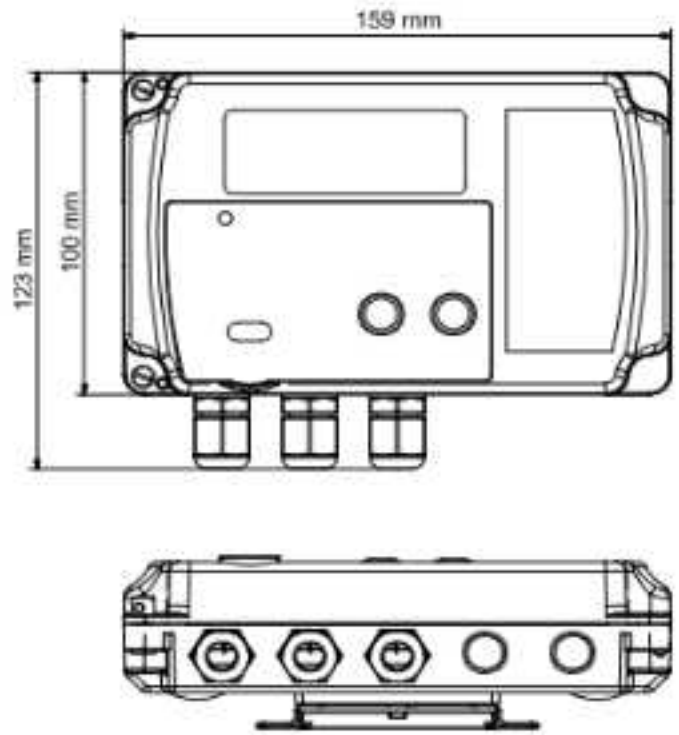


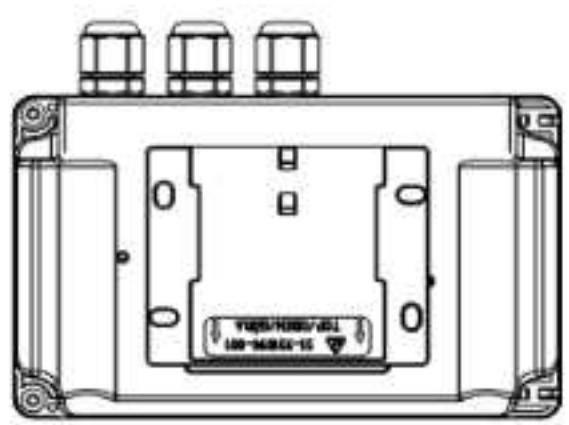

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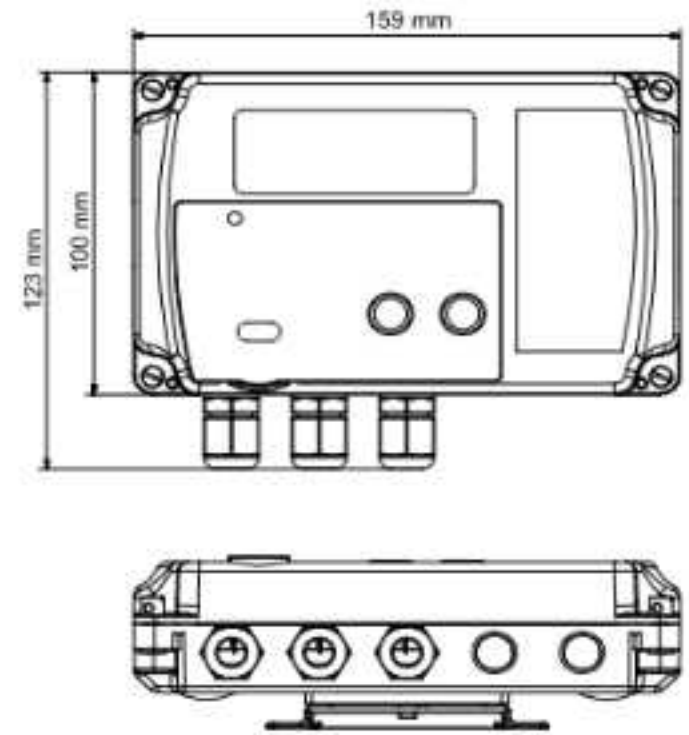
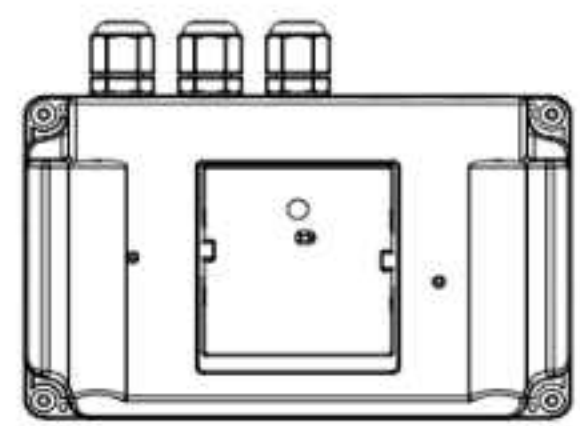
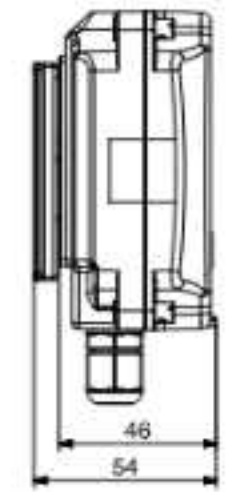
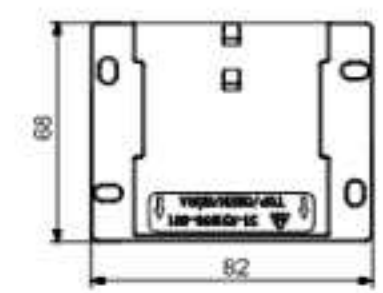




# Installation of the calculator

## Installation of the calculator on the wall

1. Remove the A mounting base (Fig. 1) from the calculator by moving it in the direction opposite than indicated by the arrows on it.
2. Mark the holes for mounting studs on the wall according to the dimensions shown in Fig. 2 or by holding the mounting base to the wall.
3. Drill the holes for the mounting studs using ø6 drill applicable to the wall material.
4. Place the mounting studs included in the set in the drilled holes.
5. Hold the A mounting base to the wall in such a way that the holes in the base match the mounting studs in the holes.
6. Attach the base to the wall using the screws included in the set.
7. Slide the calculator onto the installed mounting base as shown in Fig. 3.
8. Unscrew the B screws holding the calculator cover (Fig. 4).
9. Remove the C cover of the calculator (Fig. 5).
10. Connect electrical wires as described in the section „Electrical connections."
11. Place the C cover back on the mounting base of the calculator.
12. Screw the C cover to the mounting base using B screws.
13. Place the seals (Fig. 6 – wire, Fig. 7 – sticker).

## Installation of the calculator on the ultrasonic flow sensor

### Sharky 473 ultrasonic flow sensor

1. Remove the A mounting base (Fig. 1) from the calculator by moving it in the direction opposite than indicated by the arrows on it.
2. Place the calculator on the mounting slot of ultrasonic sensor (Fig. 8).
3. Move the calculator in the direction of cable holes until it clicks on the flow sensor.
4. Follow the instructions in points 8-13 in case of mounting the calculator on the wall.

### Ultraflow flow sensor

1. Remove the A mounting base (Fig. 1) from the calculator by moving it in the direction opposite than indicated by the arrows on it.
2. Place the mounting base on the mounting slot of ultrasonic flow sensor.
3. Move the mounting base bracket in the direction opposite than indicated by the arrows on it until it clicks on the flow sensor.
4. Follow the instructions in points 7-13 in case of mounting the calculator on the wall.

Fig. 1.

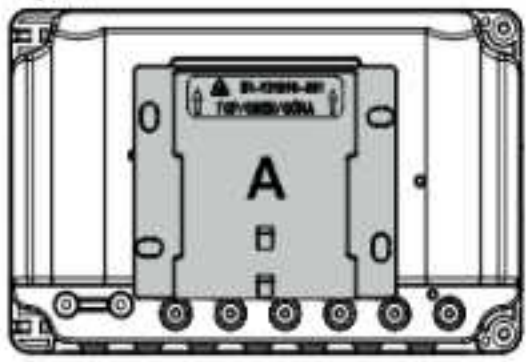



Fig. 2.

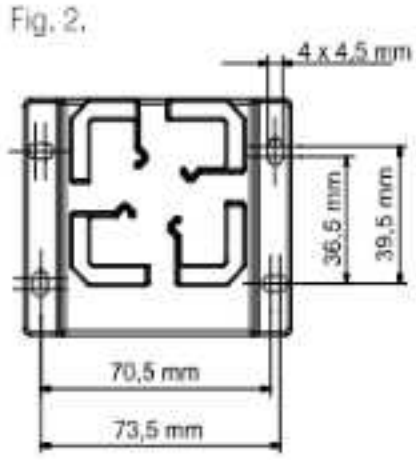



Fig. 3.

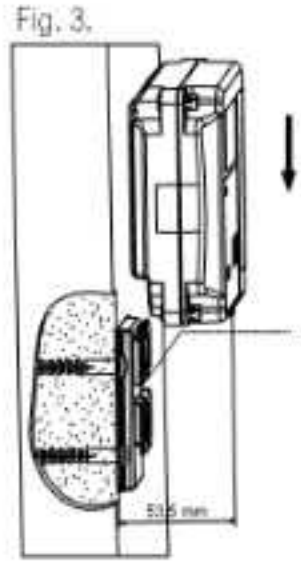


Fig. 4.

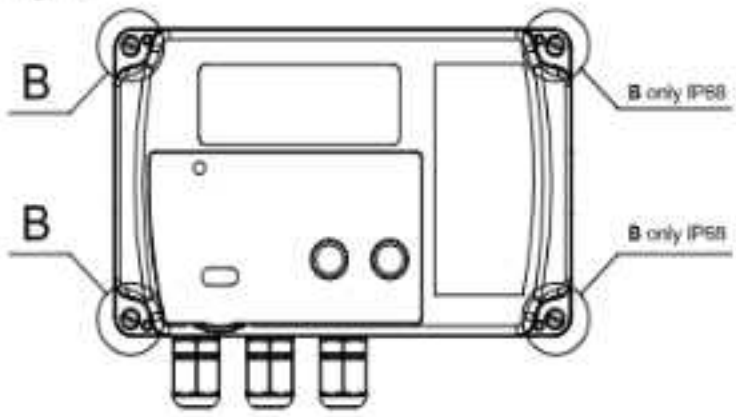



Fig. 5.

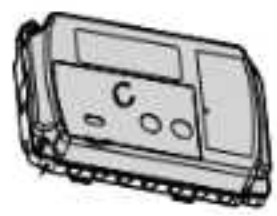



Fig. 6.

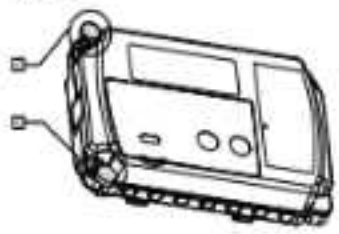


Fig. 7.

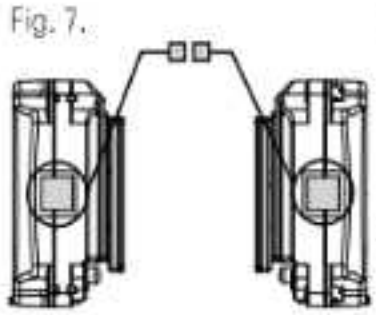


Fig. 8.

## Electrical connections

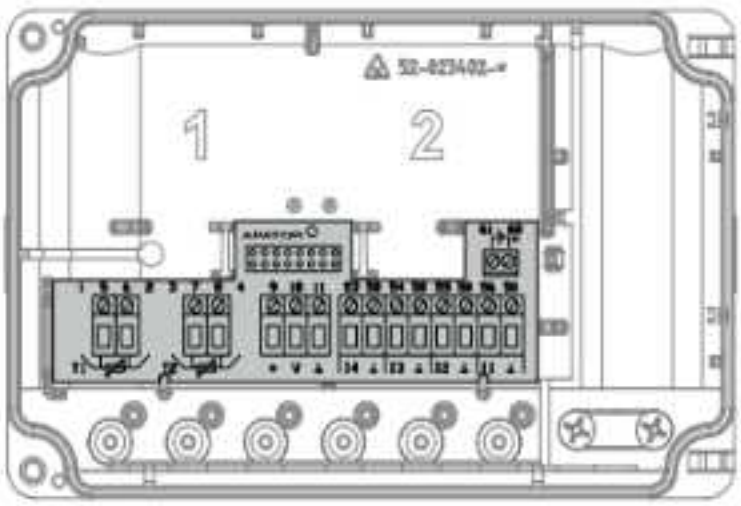
To make electrical connections, remove the C cover of the calculator (Fig. 5). The removal method is described in the section „Installation of the calculator."

### Connecting the temperature sensors

The calculator can work with three types of temperature sensors (Pt100, Pt500 and Pt1000), but for each of them there is a separate design of the main board of the calculator.

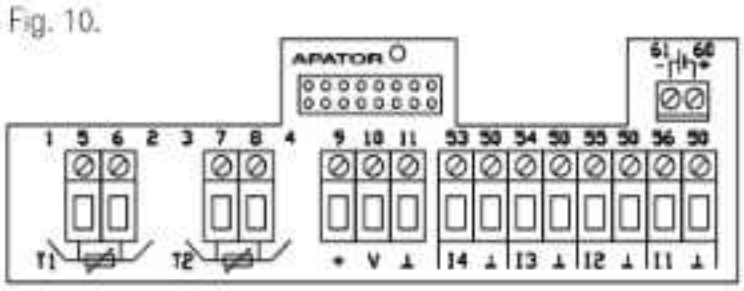
Temperature measurement can be performed using the Pt100, Pt500 or Pt1000 sensors, both 2- and 4-wire. Depending on the number of temperature sensor connection wires, the calculator connection strips are available in two versions (Fig. 9):

Fig. 9.



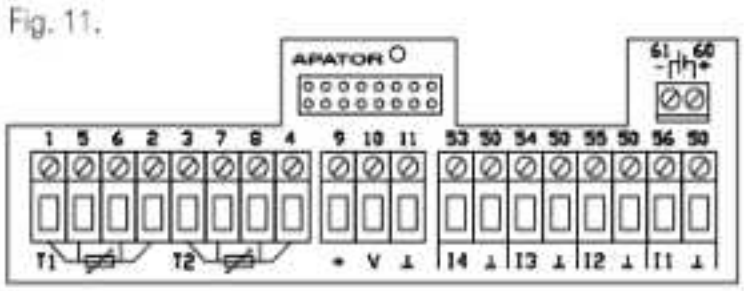

- calculator with a strip for 2-wire temperature sensor, a strip for the connection of the main flow sensor and a strip for the connection of 4 additional devices (Fig. 10),

Fig. 10.




- calculator with a strip for 4-wire temperature sensor, a strip for the connection of the main flow sensor and a strip for the connection of 4 additional devices (Fig. 11).

Fig. 11.




When measuring with 2-wire sensors (Fig. 10), the T1 sensor (power supply temperature) must be connected to the connector marked with numbers 5, 6, whereas the T2 sensor (return temperature) – to connector no. 7, 8. When measuring with 4-wire sensors (Fig. 11), the T1 sensor must be connected to the connector marked with numbers 5, 6 and 1, 2, whereas the T2 sensor – to connector no. 7, 8 and 3, 4.

### Connecting the flow sensor

The connection of the main flow sensor is made through a 3-terminal connector, marked with numbers:

9 – power supply output for flow sensor (from the main power source),
10 – signal input for the flow sensor,
11 – signal reference input for the flow sensor.

What is more, there is a possibility of connecting the flow sensor communication cable to the signal input – the fourth additional input (53).

Fig. 12.

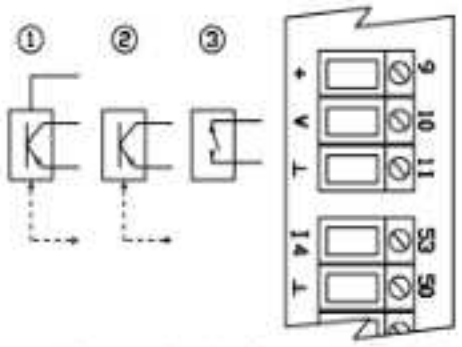
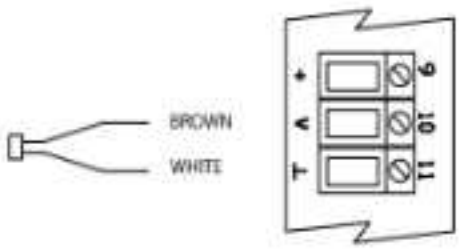



Fig. 12 shows the method of connecting flow sensors with open collector output for the sensor requiring power supply from the calculator (1), open collector output (2) and normally open contact output (3).

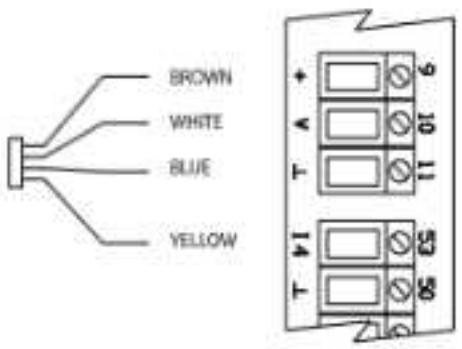
**Connecting the flow sencor with NC transmitter** - connection polarity arbitrary (Fig. 13)

Fig. 13.



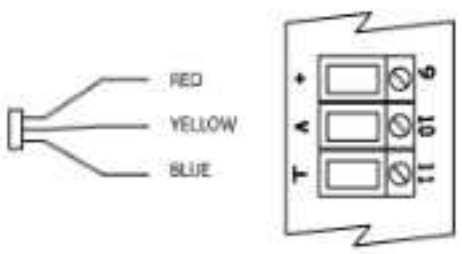

**Connecting Sharky 473 ultrasonic flow sensor (Fig. 14)**

Fig. 14.




**Connecting Ultraflow ultrasonic flow sensor (Fig. 15)**

Fig. 15.



## Connecting the additional flow sensor

The additional flow sensor may be used to detect leaks of the systems operating in the closed system. It should be connected to the additional input 3. The sensor should be connected with the 2 terminal connector marked with the following numbers:

54 - signal input for the additional flow sensor,

50 - additional flow sensor signal reference output.

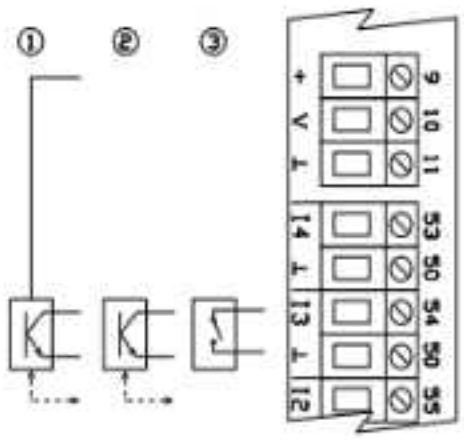
Moreover, you can also connect the sensor communications and power supply input:

9 - power supply output for the flow sensor (from the main power supply source),

55 - signal input to communicate with the additional flow sensor (additional input 2).

Please refer to the diagram below (fig. 16) to connect the additional flow sensor with the open collector output for the sensor with the required power supplied from the calculator (1), open collector output(2) and the normally closed contact output (3)

Fig. 16.




## Connecting external power supply

The calculator can be powered from 230VAC or 24VAC power adapter, as well as using AA batteries, two AA or C batteries mounted on a separate plate or D battery mounted directly in the base. Opening the cover provides direct access to the main battery or power adapter (Fig. 17). If you are replacing the D battery on your own, pay attention to how it is mounted in the casing, the positive pole should be facing the upper edge of the casing. Each battery or power adapter has cables to be connected to the calculator. Main power supply must be connected on the terminal block (Fig. 9) to the connector marked with numbers: 60 – positive pole, 61 – negative pole.

In case of calculator equipped with power supply: the power adapter should be connected to the ~230 V power supply and to the terminals marked with numbers 27 and 28 (Fig. 18); for the 24 V power adapter the ~24 V power should be supplied to terminals marked with numbers 97, 98 (Fig. 19).

The power adapter has two sets of terminals marked with numbers 95 and 96, allowing for the supply of power to interchangeable communication modules (terminals on modules marked with numbers 97 and 98). The electrical connection of power adapter should be performed by a properly licensed person.

Fig. 17.

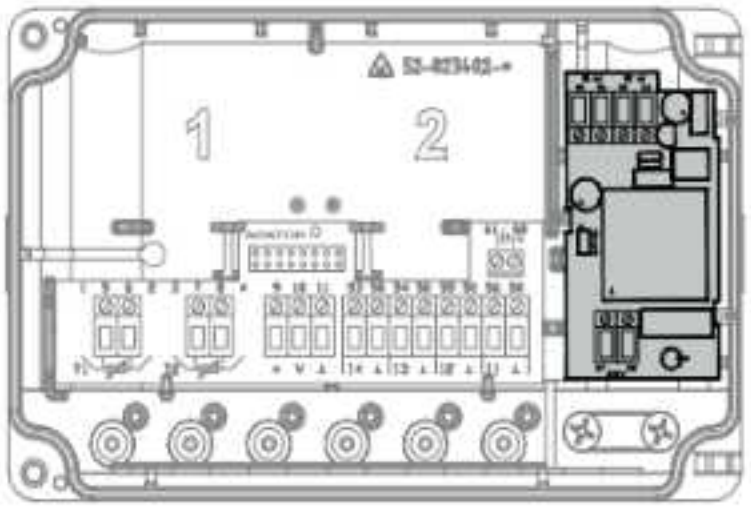



Fig. 18.

Fig. 19.

## Connecting additional input signals

Each version of the terminal block has 4 additional inputs. Each of the inputs marked I1 (terminal no. 56 and 50), I2 (terminal no. 55 and 50), I3 (terminal no. 54 and 50), I4 (terminal no. 53 and 50) can operate as an impulse input, the I3 input can additionally serve as an alarm input or additional flow sensor conection, and the I2 and I4 inputs can be used for digital communication with the flow sensor.

Each input has two terminals, marked as follows:

- signal input (terminal no. 56, 55, 54, 53)
- reference input for additional input, each input has a separate terminal for reference signal connection (terminal no. 50).

Fig. 20.

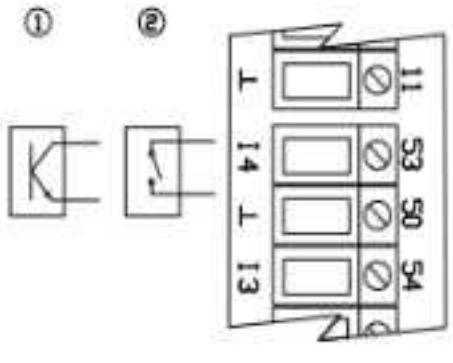



Fig. 20 shows the signal connection method for a sample additional input, for a device with output type: open collector (1), normally open contact (2).

## Installation of additional modules

The calculator allows you to install up to two independent additional modules:

- M-Bus
- RS232
- RS485
- impulse outputs (2x OB, OC or OD class output)
- impulse inputs and outputs (2x OB, OC or OD class output and 2x IB or IC class input)
- analogue outputs (2x 4-20 mA or 0-10 V output)
- LonWorks
- RF module for IMR telemetry systems (AIUT)
- Wireless M-Bus RF module.

Fig. 21.

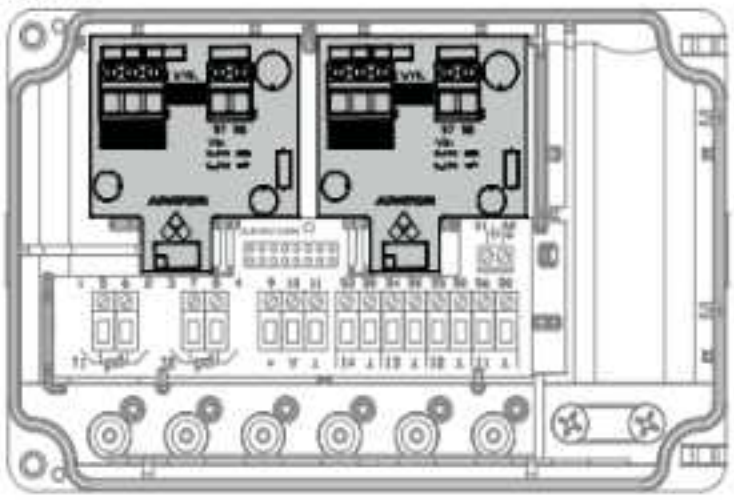


Fig. 21 shows the place and method of additional module installation. Modules can be installed in any connector except for RF modules, which can only be fitted in the connector marked with number 1 (Fig. 9).

## Installation of the M-Bus module (module code: 001)

M-Bus module is powered from the M-Bus network and galvanically isolated from the calculator using optical isolation. The module supports primary, secondary and extended secondary addressing. The M-Bus signals are connected through the inputs marked with numbers 24, 25 (Fig. 22).

## Installation of the RS 232 module (module code: 003)

The RS 232 module requires an external power source. It is galvanically isolated from the calculator using optical isolation. The module can be powered by AC or DC voltage. The power source is connected through terminals no. 97, 98, the polarity of the connected power supply is arbitrary. The communication cables are connected through the following terminals: 64 – signal ground, 63 – TxD module output, 64 – RxD module input (Fig. 23).

## Installation of the RS 485 module (module code: 003)

The RS 485 module requires an external power source. It is galvanically isolated from the calculator using optical isolation. The module can be powered by AC or DC voltage. The power source is connected through terminals no. 97, 98, the polarity of the connected power supply is arbitrary. The communication cables are connected through the following terminals: 86 – signal ground, 85 – A+ input/output (non-inverting), 85 – B- input/output (inverting) (Fig. 24).

## Installation of the LonWorks module (module code: 002)

The LonWorks module (Fig. 25) requires an external power source. The module can be supplied with direct current at a voltage range of 9-24 V. Power source is connected by means of a pair of terminals located in the upper right part of the module designated as B. The positive terminal is marked as Vcc, whereas the negative pole – as GDN. Communication cables are connected by means of a pair of terminals located in the upper left part of the module designated as A – the polarity is arbitrary.

## Installation of the Wireless M-Bus module (module code: 004)

The Wireless M-Bus module (Fig. 26) is battery-powered and has its own antenna, so for the purpose of proper operation of the module, just put it in the appropriate connector.

**NOTE!** Wireless M-Bus modules can be mounted only in the connector marked with number 1 (Fig. 9).

Fig. 22.

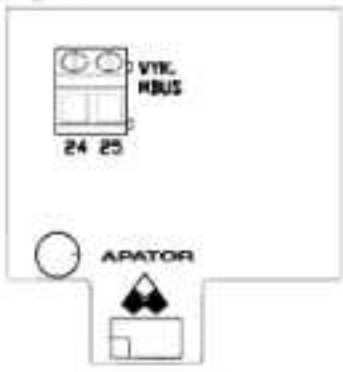



Fig. 23.

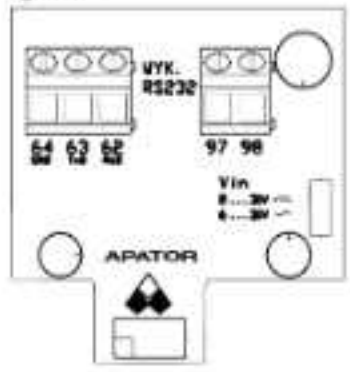



Fig. 24.

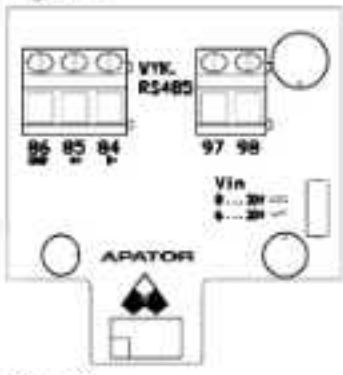



Fig. 25.

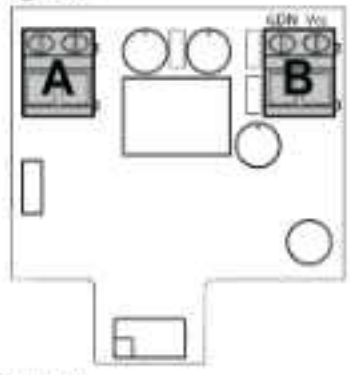



Fig. 26.

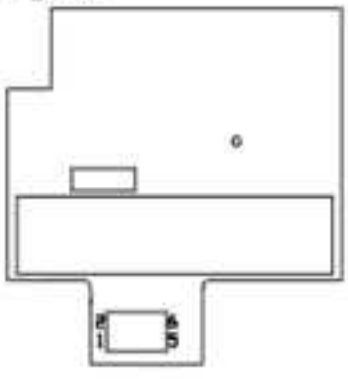


Fig. 27.

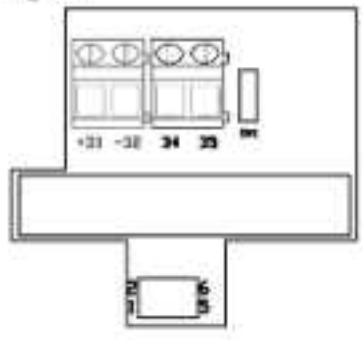


Fig. 28.

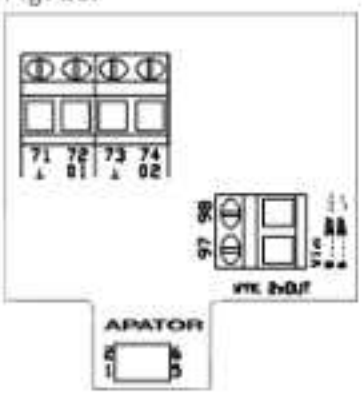
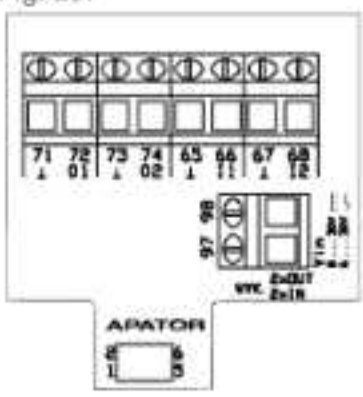



Fig. 29.




## Installation of IMR module (module code: 005)

There are three types of IMR modules (Fig. 27): with radio interface (APAT 1312), with radio and cable interface (APAT 1322), with cable interface (APAT 1332). Modules can be powered, depending on the version, from their own battery or external power source. Cable interface (terminal no. 34 and 35) can be used to connect an active AMPLI antenna – cable order does not matter. For the connection of external power source (in the range of 4-16 V or stabilised 3.6 VDC) the following terminals are used: 31 – positive pole, 32 – negative pole. The module supplied by the manufacturer with its own power supply is in the sleep mode. The APAT 13x2 module can be activated using the SW1 diagnostic button (push and hold the button for 5 to 7 seconds, then let go). If APAT 13x2 is a version without its own power supply or if the system is awake, push and hold the diagnostic button for 5 seconds to check the operating mode. Correct operation is signalled with 4x flash of the LED.
NOTE! IMR modules can be mounted only in the connector marked with number 1 (Fig. 9).

## Installation of the impulse input/output module

There are two types of the pulse input/output modules:
1) 2 pulse outputs module (module code 016), the module is shown in figure 28.
2) 2 pulse inputs and 2 pulse outputs module (module code 080), the module is shown in figure 29.

Depending on the version the modules can be supplied from the Faun calculator or the external power source. The external power source is connected to terminals 97, 98. The polarisation of the connected power can be any. Figures 28 and 29 present the module supplied from the external power source. When the module is supplied from the calculator it is not equipped with the power terminal. Due to the high average current consumption by the pulse outputs it is recommended to use modules supplied from the external power source.
The pulse inputs are connected to terminals 65, 66 and 67, 68 and the signal from the pulse outputs is connected to terminals 71, 72 and 73, 74.
The pulse outputs can operate in one of three classes: OB, OC or OD (according to EN 1434-2). The output operation class is configured during production and

Fig. 30.

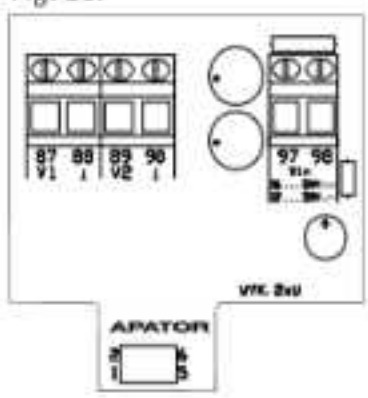
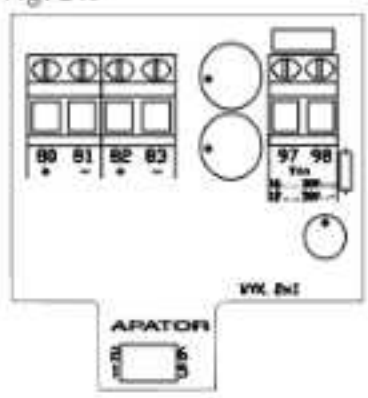



Fig. 31.




cannot be subsequently changed. The individual pulse output classes have different maximum pulse frequency, pulse duration and interval.
The pulse output generates the number of pulses which is proportional to the increase in (main, cool or tariff) energy or volume with the selected pulse value. The type of the pulsed data and the pulse value can be configured in any manner for each output.

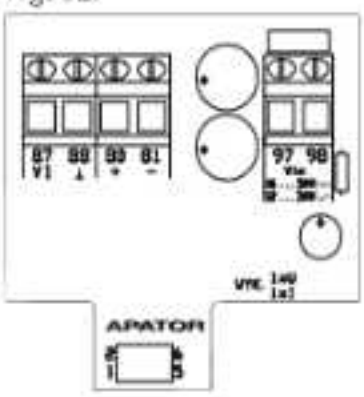
## Installation of 2 analogue outputs module (module code: 256)

The analogue output module requires an external source of power. There is a galvanic isolation between the calculator and the module. The module is available in three versions with two current outputs 0/4 ... 20 mA, two voltage outputs 1 ... 10V or one current output and one voltage output. You can freely configure the analogue outputs in such a way so as to output a status proportionate to one of the momentary values: flow, power, supply temperature, return temperature or temperature difference. The status of the analogue output is updated every cycle when momentary values are determined, i.e. every 10 seconds during normal operation with flow other than zero. For correct operation, it is necessary to adjust the outputs in the module configuration.

Terminals 97 and 98 are to connect power supply of the module. Analogue output terminals, depending on workmanship, are described in the following way, for the workmanship of:
1) 2 voltage outputs (Fig. 30):
   - output 1 – terminal 87, reference to output 1 – terminal 88,
   - output 2 – terminal 89, reference to output 2 – terminal 90.
2) 2 current outputs (Fig. 31):
   - output 1 positive terminal – 80, negative terminal – 81,
   - output 2 positive terminal – 82, negative terminal – 83.
3) 1 current output + 1 voltage output (Fig. 32):
   - output 1 (voltage output) – terminal 87, reference to output 1 – terminal 88,
   - output 2 (current output) positive terminal – 80, negative terminal – 81.

**NOTE:** When the module is powered from calculator power supply, due to max. load-carrying ability of power supply, it is possible to install one analogue module only.

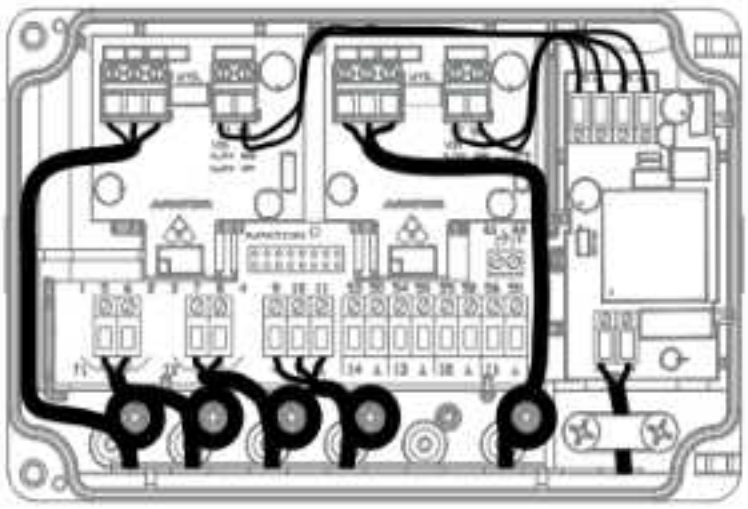
Fig. 32.




## Cable lining inside the calculator

The calculator has been designed to assist you in the installation process as much as possible. In the base of the device there are special "tunnels" that help in connection cable lining. The IP54 version of the calculator also features special pins preventing the cable from being ripped out. In IP65 and IP68 models this task is performed by cable grommets. Figure 35 shows how cable is to be lined inside the device.

Fig. 33.



## LCD screen

Fig. 34.

Description of the symbols displayed on the calculator screen (Fig. 34) is presented below.

1 Symbol indicating the display of volume or flow value, as well as additional input values when displayed along with the symbol in the field no. 2.
2 Symbol of additional input or tariff, indicating the display of data (register or configuration value) for additional input, symbols: A, b, C, d or the values associated with tariffs, symbols: 1, 2.
3 Symbol of instantaneous/minimum/maximum/average or tariff value, indicating the display of one of the following values:
- instantaneous value: , for the flow and power,
- minimum value: ,
- maximum value: ,
- average value: ,
- tariff value: , displayed together with the symbol in the field no. 2, indicating the tariff number.
4 Symbol of power supply temperature, displayed for the instantaneous, average or peak value, as well as archived power supply temperature.
5 Symbol of temperature difference, indicating together with symbols 4 and 6 the display of temperature difference.
6 Symbol of return temperature.
7 Symbol of cold, displayed for the valuesof cold registers: energy and volume, as well as on the screen of temperature difference measured in time, when the conditions of cold measurement are met.
8 Symbol of failure, indicating the occurrence of failure. The symbol is displayed on the main screen and on the screens associated with that failure, where in the field no. 18 a message on error type is additionally displayed.
9 Symbol of time, displayed on date and current time screens for the date and time values associated with archived data and working time.
10 Symbol indicating the access level to the configuration of calculator (displayed on all screens during access unlocking). The blinking symbol indicates the access to metrological settings, whereas steady light symbol – access to user settings.
11 Symbol of return flow, indicating the flow at variance with the proper direction. Available only for flow sensors with digital communication.
12 Symbol of flow, indicating flow in the proper direction.
13 Symbol of low battery, displayed along with the symbols 14 and 15, indicates the type of low battery (displayed on all screens during the occurrence of failure) or the type of battery, for which the voltage is displayed in the service menu.
14 Symbol of main battery, displayed along with the symbol 13, indicates the values or signals the failure of the main battery.
15 Symbol of backup battery, displayed along with the symbol 13, indicates the values or signals the failure of the backup battery.
16 The main display field, 8-digit field for displaying valuesand configuration data of the calculator.
17 Symbol of the archive, displayed with the symbol 19, 21, 22, 23, 24 or 25, indicates the type of archived data displayed.
18 Additional display field, 4-digit field for displaying additional descriptions of the valuesdisplayed in the main field, the error type and the current or archived time on selected screens.
19 Symbol of daily data, indicating the display of archived data from the daily archive.
20 Symbol of optical port activity, appears when the optical port is active and communication is possible.
21 Symbol of monthly data, indicating the display of data from the monthly archive.

22 Symbol of annual data, indicating the display of data from the annual archive.
23 Symbol of hourly data, indicating the display of archived data from the hourly archive.
24 Symbol of minute-based data, indicating the display of data from the minute-based/configurable archive.
25 Symbol of tariff data, indicating the display of data from the tariff archive.
26 Decimal symbol, indicating the decimal part of a value or separating different values.
27 Decimal separator symbol, highlighting the decimal part of a value.
28 Symbol of impulse frequency, indicating the impulse frequency unit.
29 Unit field, indicating the unit of value displayed in the main field, available units of energy, volume, flow, power and time.
30. Register type symbol. Displayed only when the calculator is configured to operate in the leak detection system, only for the values determined for both sensors (volume, weight, flow and error code). It specifies the flow sensor which the displayed data relates to: A - main flow sensor, B - additional sensor.

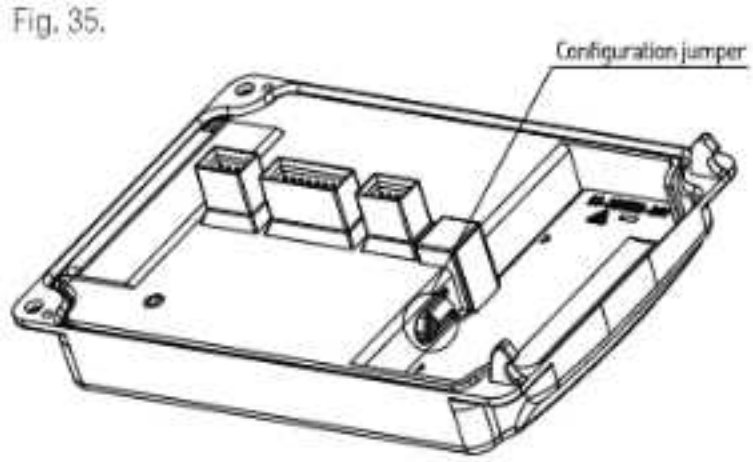
## Error code description

| Code | Description | Time |
|---|---|---|
| 1 | Long-time leak | conf. |
| 2 | No flow, ΔT difference correct | conf. |
| 4 | Error: supply temp. sensor | < 10 s |
| 8 | Error: return temp. sensor | < 10 s |
| 16 | Error: reverse temperature difference | < 10 s |
| 32 | Error: (maximum) flow exceeded | < 10 s |
| 64 | Low main battery or no mains power | < 10 s / < 24 h |
| 128 | Low backup battery | < 24 h |
| 256 | Additional input alarm | < 1 s |
| 512 | Wrong flow direction | < 3 min. |
| 1024 | Air in flow meter | < 3 min. |
| 2048 | Flow meter's measuring system damaged | < 3 min. |
| 4096 | Error: no communication with flow sensor | < 3 min. |
| 8192 | "Short-time" leak | conf. |
| 16384 | Improper module 1 | < 10 s |
| 32768 | Improper module 2 | < 10 s |

## Configuration of the calculator from the menu

The calculator allows for a manual configuration of selected parameters from the menu. You can set parameters such as date, time, network settings and additional input settings. Configuration possibility is secured by a configuration jumper available after opening the casing (Fig. 35).
Upon pressing and holding the jumper for 1 second, the configuration group selection screen is automatically displayed (06). Configuration group is available for 5 minutes after using the jumper or after recent activity in this group. After this time, the display automatically returns from the configuration group to the main energy; to enable configuration group, use the jumper again. When the possibility of configuration is active, at any time it is possible to exit and re-enter the group in a standard way. Upon moving to the configuration menu the current calculator configuration is read and then displayed during editing.

Fig. 35.




Navigating between screens of the configuration group takes place in a standard way.

Fig. 36.

### Numeric value editing

To start editing a numeric value, briefly push the P2 button at the time of its display (Fig. 36). The first digit of the edited value will start flashing. By briefly pushing the P2 button, set the appropriate value; to move to the next digit editing, briefly push the P1 button (Fig. 36). To exit the editing mode of a given value, go to the last editable digit and briefly push the P1 button once more. As a result, the last digit will stop flashing and the screen will return to the value display mode, after which it will be possible to select the next value to edit.
In this way it is possible to configure the following parameters in the calculator:
- date and time,
- network address and customer ID,
- impulse frequency and serial number for additional inputs
- initial value of the meter for additional inputs.

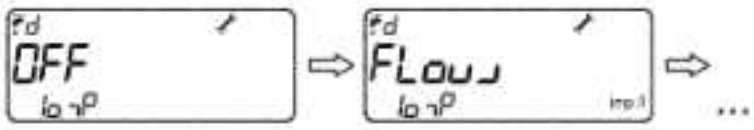
### Date and time editing

Configuration of date and time takes place on the same menu screen. To edit the date, briefly push the P2 button when in the settings menu. Editing begins with the year, each next push of the P1 button enables the editing of the month, day, hour and minutes. Set date

and time are saved in the calculator upon finishing the editing of the last digit (tens of minutes).

## Additional input type editing

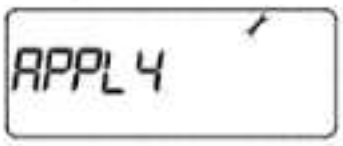
To change the type of the additional input, briefly push the P2 button when displaying the impulse input type screen. As a result, the input type will be changed to another type available. Depending on the input number, it is possible to set the following types of inputs:
– inactive input,
– impulse input with impulse frequency: $dm^3$/imp, imp/$dm^3$, imp/kWh,
– alarm input,
– input for digital communication with sensor.




**NOTE:** In case of setting the output type other than an impulse one in the menu, the impulse frequency and serial number settings for the given input are not displayed.

## Confirmation of the edited parameters

When setting date and time, the parametersin the calculator are updated immediately after the end of editing.




In order to confirm the remaining values, go to the last screen – confirmation of configuration group changes. If there have been changes made to the current configuration, the APPLY option will appear on the screen. To save the changed configuration, push and hold the P2 button until the screen displays „No rConF". This means that the calculator is already configured in accordance with the data entered in the configuration menu.

**NOTE:** If the configuration group is abandoned without the implementation of the operation described above, introduced settings will not be saved in the calculator.

## Default configuration of the calculator

- **Additional inputs** 1-4 set as impulse inputs with an impulse frequency of 1 l/imp (communication with the sensor inactive).
- **Annual logs:** 24 with average, maximum and minimum values.
- **Monthly logs**: 172 with average, maximum and minimum values.
- **Daily logs**: 662 with average and maximum values.
- **Hourly logs:** 1236 with instantaneous values.
- **Minute-based logs**: 1777, 5 min period, with instantaneous values.
- **Billing logs**: 160.
- **Tariff logs**: 668.
- **Maximum flow exceeded error:** inactive.
- **No flow error parameters**: 48h, 10°C, flow threshold 0, active error.
- **Delta T**: 0,1°C.
- **Protocol in both modules**: M-Bus, 2400, even.
- **Averaging period**: 60 min
- **Transmission speed for optical port:** 9600 bps.
- **Tariffs:** inactive.
- **Measurement of cold:** inactive.
- **Negative difference temperature error:** active.
- **The maximum number of archived records displayed:**
  – tariff: 40,
  – minute-based: 120,
  – hourly: 48,
  – daily: 62,
  – monthly: 36,
  – annual: 6,
  – billing: 12.
- **Time of saving to the daily, monthly, annual archive:** hour: 1, day: 1, month: 7.
- **Time of saving to the billing archive:** hour: 1, day: 15, monthly basis.
- **Time of return to the main menu indication:** 120 s.
- **Display ranges for the main menu and statistical group:** full.
- **Network no.**: 1.
- **Customer number:** serial number.

The above configuration is applicable unless at the time of ordering no other specification was provided.

## Display sequence

### ---- 01

- Energy
- Cool energy
- Energy tariff 1
- Energy tariff 2
- Main volume
- Cool volume
- Volume tariff 1
- Volume tariff 2
- Mass
- Power supply temperature
- Return temperature
- Temperature difference
- Instantaneous main flow
- Instantaneous additional flow
- Instantaneous power
- Error code
- Input 1
- Input 2
- Input 3
- Input 4
- Metrological test
- Display test

### ---- 02

- Average, max., min. flow
- Average, max., min. power
- Average, max., min. power supply temperature
- Average, max., min. return temperature
- Average, max., min. temperature difference

### ---- 03

**Meter**
- Serial number
- Customer number
- Network address
- Main impulse weight
- Installation place, operation type
- Date and time
- Program version
- Production date
- Operating time
- Operating time with error
- Cooling energy measurement threshold
- Time of saving in billing archive
- Error threshold for exceeded flow
- Error threshold for no flow
- Battery voltage

**I/O**
- Configuration of additional input 1, 2, 3, 4
- Type and configuration of installed communication modules
- Optical connector configuration

**Regs**
- More recent numbers: energy, additional energy, tariff 1 energy, tariff 2 energy

### ---- 04

- Energy tariff 1
- Volume tariff 1
- Operating time at tariff 1
- Threshold type tariff 1
- Threshold value tariff 1
- Energy tariff 2
- Volume tariff 2
- Operating time at tariff 2
- Threshold type tariff 2
- Threshold value tariff 2
- **Tariff archive**
  - Record no.
  - Threshold occurrence time
  - Threshold decline time
  - Tariff energy
  - Tariff volume
  - Operating time at a tariff

### ---- 05

**Type of archive**
- Record no.
- Date and time of record
- Energy main, cool
- Energy tariff 1, 2
- Volume main, additional
- Volume tariff 1, 2
- Volume cool
- Mass main, additional
- Error code
- Instantaneous: flow, power, power supply temp., return temp., temp. difference
- Flow (average, max., max. occurrence time, min., min. occurrence time)
- Power (average, max., max. occurrence time, min., min. occurrence time)
- Power supply temperature (average, max., max. occurrence time, min., min. occurrence time)
- Return temperature (average, max., max. occurrence time, min., min. occurrence time)
- Temperature difference (average, max., max. occurrence time, min., min. occurrence time)
- Input 1, 2, 3, 4
- Operating time
- Operating time with error

### ---- 06

- Year, month, day, hour, minute
- Customer number
- Network address
- Type: input 1, 2, 3, 4
- Impulse frequency: input 1, 2, 3, 4
- Initial state: input 1, 2, 3, 4
- Serial number: input 1, 2, 3, 4
- Confirmation of changes made

IOTAFLOW
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# ULTRASONIC BTU METER MANUAL


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